

Gene, Transcript, Protein, and Perturbation Identifier Systems

A first-principles monograph for the MANTRA perturbation-response data stack

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Abstract

A biological experiment uses several kinds of identifiers. Gene symbols name genes for readers. Transcript and protein accessions identify particular database records. Guide IDs identify experimental constructs. Model keys identify rows and columns in data arrays. These identifiers are connected, and each has a specific job.

This monograph explains the biology behind those distinctions, how genome annotation represents genes and their products, and how major databases identify the resulting records. It then follows CRISPR guide designs through single-cell sequencing to the MANTRA prediction matrices.

The central design rule is to preserve each identifier with its object type, source, and release, then record the relationships used to connect it to other objects. For MANTRA, this means keeping historical intervention labels, resolved biological records, and ordered response-column keys separately identifiable.

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1 The Biological Identity Problem

1.1 Why a gene-identification system becomes difficult in practice

A gene name tells a reader which gene is being discussed. A data pipeline needs a more specific answer: what does each identifier name, and how does that object connect to the experiment?

Consider an experiment that reduces the activity of a gene and measures the resulting changes in RNA abundance. The experiment involves a target gene, a reagent used to alter its activity, and a table of measurements. A later analysis also retrieves information about the gene's transcripts and proteins. Each related object needs its own identifier.

For example, TP53 is a human-readable gene symbol. ENSG00000141510 identifies an Ensembl gene record. A transcript identifier points to a particular RNA structure associated with that gene. A UniProt accession (database identifier) points to a protein database entry. A guide-pair identifier names the experimental construct used to deliver two targeting guide RNAs. Each identifier answers a different question.

The measurement table adds another distinction. A column number says where a measurement is stored. The gene identifier says what that column represents. Moving the column changes its position; the gene stays the same.

Worked example: one experiment, four questions

Question	Object being identified
Which gene is the intended target?	A gene, represented by a gene record
What intervention was delivered?	A particular guide construct
Which gene's RNA was measured?	The gene assigned to a measurement column
Where is that measurement stored?	A column in a particular data table

The target gene and a measured gene play different roles. Reducing one gene's activity may change the abundance of RNA from many other genes. The experiment therefore needs both a target identity and a separate list of measured gene identities.

Names also change over time. If a library used an older gene symbol, preserve that experimental label and record its mapping to the current gene record. This keeps the original experiment interpretable while allowing newer databases to be used.

MANTRA stores these identifiers in separate fields and records their connections. The chapters that follow explain the biology behind those connections and the databases that name each object.

1.2 Four categories of objects

Use the following four categories to distinguish what a record is describing. Classify an object by its role in the analysis. A physical cell, for example, is biological; its sample label belongs to the experiment, and its measurement row belongs to the data table.

Category	What belongs here	Example
Biological entities	The molecules and genomic regions being studied	A gene's DNA region; an RNA molecule; a protein molecule
Database records	Published descriptions of those entities	An Ensembl transcript record
Experimental designs and observations	Interventions, sample identities, and observations defined by an experiment	A guide-pair design; a sample label
Computational positions and keys	The organization of values inside a dataset or model	An expression-table column; a response-axis key

A connection links two distinct objects. An RNA molecule exists in a cell. A transcript record describes the sequence and structure used to identify such molecules. An experiment measures RNA, and a data table stores the resulting numbers.

1.3 A typed model of identity

Store an identifier with enough context to interpret it. A **namespace** is the system that issues an identifier. An **object type** states what the identifier names. An **accession** is a database's key for a record; a project-defined identifier is a local key.

Field	Question it answers
Namespace	Who issued this identifier: Ensembl, HGNC, or MANTRA?
Object type	Does it name a gene record, transcript record, guide pair, or model coordinate?
Accession or local key	Which particular record or object?
Version	Which revision of this individual record or sequence?
Release or snapshot	Which published collection of records was used?
Genome assembly	Which reference DNA sequences define its coordinates?
Provenance	Which source file or procedure supplied this information?

Together these seven fields form the document's proposed identity record. The compact notation in later discussions is $I = (n, o, a, v, r, b, p)$, in the order shown above. Version and assembly apply only where the issuing system uses them.

For example, ENST00000611639.4 contains an Ensembl transcript accession and version 4. The annotation release and source file are additional information. Record that context beside the accession. A source-file hash, a value computed from the file's bytes, identifies the exact file used.

1.4 Orthogonality as a database-design concept

The resources in this monograph answer different questions about a biological experiment. Here, **orthogonality** means keeping those questions separate in the data model. The databases share information and depend on one another.

Resource	Main question it answers here
HGNC	What is this human gene's approved name and symbol?
Ensembl	Which gene or transcript feature does this accession identify?
GENCODE	Which gene and transcript structures are included in this annotation release?
RefSeq	Which reference RNA or protein sequence does this accession identify?
UniProt	What protein sequence and protein knowledge does this entry describe?
ENCODE	Which functional-genomics experiment or data file supplied this measurement?
MANTRA	Which intervention row or model output position does this local key identify?

A **cross-reference** links records between resources. A **crosswalk** collects those links. For example, a gene record links to its transcript records; the coding transcripts link to translated protein sequences.

The number of links depends on the relationship. In a worked example with one gene and two transcript structures, the gene-to-transcript mapping contains two links. If both transcripts encode the same amino-acid sequence, there is only one distinct protein sequence in that example. This is why a program must name the relationship it is following before treating a mapping as one-to-one.

Record the source release for each mapping. If an annotation release adds a transcript to a gene, the gene-to-transcript mapping changes even when the gene accession stays the same.

Three fields worth keeping separate. A symbol such as TP53 is a readable label. An accession such as ENSG00000141510 identifies a database record. A coordinate such as response column 1,642 identifies a position in a particular model output. The column number here is illustrative.

1.5 Scope of this monograph

Chapter 2 explains the biological objects: DNA, genes, RNA transcripts, proteins, and their relationships. Chapter 3 explains how an annotation represents those objects on a reference genome. Chapter 4 describes the identifier systems used to name them.

Chapter 5 explains CRISPR interventions and guide designs. Chapter 6 follows RNA through single-cell sequencing into a data matrix. Chapters 7 through 9 describe the MANTRA identifier contract and its use. The appendices provide reference tables, a glossary, and sources.

2 Biological Foundations

2.1 DNA as a physical molecule and an information-bearing polymer

DNA is a long molecule built from repeating units called **nucleotides**. Each nucleotide contains a sugar, a phosphate group, and one of four bases: adenine (A), cytosine (C), guanine (G), or thymine (T). The order of these bases is the DNA sequence.

In the DNA double helix, two strands run in opposite directions. A pairs with T, and C pairs with G. These pairing rules let a cell use one strand as a template for making a complementary strand.

The ends of a strand are called **5-prime (5')** and **3-prime (3')**, after numbered positions in its sugar. Enzymes called polymerases build a new nucleic-acid strand in the 5'-to-3' direction. A gene's strand tells us which direction to follow along a reference sequence when reading that gene.

Human nuclear DNA is organized into chromosomes. Most human body cells carry two copies of each autosome, one from each parent, together with two sex chromosomes. The inherited copies differ at some positions. A reference genome supplies shared sequences and coordinates for analysis; an individual's DNA differs at some reference positions.

Chromatin is DNA packaged with proteins. DNA wraps around histone proteins to form units called nucleosomes. This packaging influences access to the DNA. Measurements of chromatin help researchers study whether a region is accessible to the machinery that controls gene activity.

2.2 Genome, assembly, chromosome, contig, and coordinate

A **genome** is the complete genetic material of an organism or cell. A **chromosome** is a physical DNA-containing structure. A **genome assembly** is a reconstructed sequence representation built from sequencing data. A **contig** is one continuous assembled sequence; contigs are used to build longer reference sequences, including chromosome representations.

GRCh38 names a human reference assembly. GRCh38.p14 specifies its patch release. A patch provides a correction or an alternative representation of a reference region.

A **genomic coordinate** is a numbered position on a named reference sequence. A genomic interval identifies a stretch of that sequence by its start and end; strand specifies orientation. Always name the assembly because position 100,000 refers to the sequence of the particular assembly being used.

Coordinate conventions also matter. In a one-based inclusive convention, positions 1 through 3 include the first three bases. In a zero-based half-open convention, the same three bases occupy [0, 3): the start is included and the end is excluded. GTF uses the first convention; BED conventionally uses the second.

2.3 The historical and modern meanings of a gene

For the biology introduced here, a **gene** is a region of DNA whose sequence specifies an RNA product. For a protein-coding gene, that RNA carries the instructions for making a protein. For a noncoding RNA gene, the RNA is itself the product of interest.

A **locus** is a genomic location or region. A **gene record** is a database's description of a gene at a locus. The record groups the associated transcript structures and supplies identifiers and other annotations. Databases also retain records for pseudogenes, discussed in §2.13.

The meaning of gene developed historically. Early genetics used the term for a unit of inheritance, inferred from how traits passed between generations. Molecular biology connected inheritance to DNA sequences and their products. Genome annotation now gives those sequences explicit boundaries and records.

One gene can produce distinct RNA sequences through RNA processing. In the example in §2.6, the same gene produces one RNA containing exons 1-2-3-4 and another containing exons 1-3-4. Both belong to the same gene; they have different transcript identities.

2.4 Locus boundaries, promoters, enhancers, and regulatory regions

A **promoter** is the DNA region where the transcription machinery assembles to begin making RNA. The **transcription start site** is the position where that copying begins. Choosing a different promoter changes where transcription starts.

An **enhancer** is a regulatory DNA element that increases transcription from a promoter when the relevant regulatory machinery is active. Enhancers also act over long stretches of DNA. A **silencer** reduces transcription in its regulatory context. An **insulator** helps delimit regulatory interactions or chromatin domains. These labels describe regulatory roles. A region's role depends on the regulatory context.

Cis describes regulation through an element on the same DNA molecule as its target. **Trans** describes regulation by a diffusible molecule, such as a protein that binds DNA. In a concrete case, a promoter is a cis-regulatory DNA region; a transcription-factor protein binding that promoter acts in trans.

For a perturbation experiment, record the actual region targeted. A guide directed at a gene's promoter and a guide directed at a distant enhancer act at different DNA sites, even if the experiment aims to change the same gene's activity.

2.5 Transcription: synthesis of RNA from a DNA template

Transcription makes RNA by copying a DNA template. RNA polymerase begins at a transcription start site, extends the RNA strand, and eventually stops. These stages are called initiation, elongation, and termination. RNA uses U (uracil) where DNA uses T.

In the human protein-coding example developed below, the initial RNA is **pre-mRNA**. Processing converts it into mature messenger RNA (mRNA), which supplies the template for translation. Processing includes capping, splicing, and formation of the RNA's 3' end; these processes also occur while transcription is underway.

2.6 RNA processing and alternative splicing

An **exon** is a sequence segment retained in a particular mature transcript. An **intron** is a transcribed segment removed during splicing. The spliceosome, a complex of RNAs and proteins, removes introns and joins the retained exons.

Exon and protein-coding region answer different questions. Exon describes what remains after splicing. Coding region describes what the ribosome reads to make protein. An exon may include coding sequence, untranslated sequence, or both. The untranslated regions (UTRs) are the portions of an mRNA before and after its coding region.

In the usual processing pathway for human mRNA, a protective cap is added at the 5' end. The RNA is cleaved at its 3' end and a poly(A) tail, a run of A nucleotides, is added. Splicing determines which transcribed segments are joined in the mature RNA.

Alternative splicing produces different mature RNA sequences by changing which splice sites are used. Follow this illustrative gene with four numbered exons:

RNA structure	Exons retained	What changed?
Transcript A	1 - 2 - 3 - 4	All four exons are included
Transcript B	1 - 3 - 4	Exon 2 is skipped
Transcript C	1 - 2 - 4	Exon 3 is skipped

Each row is a different **transcript isoform**: one particular mature RNA structure associated with the gene. The example establishes three RNA structures. Determining their protein products requires examining each RNA's coding sequence.

A transcript's structure is described by its start and end positions and the ordered exonic intervals on its strand. Alternative starts and ends also create transcript differences while retaining the same internal splice pattern. In mutually exclusive exon use, one transcript includes one of two alternative exons and another includes the other.

A **transcript model** is the database record describing an inferred RNA structure. Researchers construct that model from observations such as sequenced RNA or cDNA. Thousands of physical RNA molecules may match one model. Each molecule is a physical copy; the model is their shared reference description.

An annotation assigns separate transcript identifiers to the distinct structures it represents. This preserves the difference between transcripts A, B, and C while linking all three to their gene.

2.6.1 Exon skipping

In exon skipping, one RNA includes an exon and another leaves it out. In the preceding example, transcript A joins exons 1-2-3-4; transcript B joins 1-3-4. The splice joins exon 1 directly to exon 3.

The protein effect depends on the removed sequence. Removing a segment entirely within a UTR leaves the encoded amino-acid sequence unchanged. Removing coding bases changes the coding sequence; whether later codons remain in the same groups of three depends on the number of bases removed.

2.6.2 Alternative splice sites

A **splice donor** marks an intron's beginning; a **splice acceptor** marks its end, following the RNA's direction. Using a nearby alternative site changes the boundary between retained and removed sequence.

For example, shifting an exon boundary by three bases changes the mature RNA by three bases. Shifting it by one base changes it by one. Within a coding region, a one-base change shifts the downstream reading frame: the ribosome groups the remaining bases into different triplets.

2.6.3 Intron retention

In intron retention, a segment removed from one transcript remains in another. Compare exon 1 joined directly to exon 2 with exon 1 joined to the intervening sequence and then exon 2. The second RNA retains that intron.

The retained sequence affects the RNA's fate. A stop codon introduced within a translated region, for example, may trigger RNA quality-control mechanisms that degrade the transcript. Retention also affects processing and export from the nucleus; its outcome depends on the transcript and cell context.

2.6.4 Alternative transcript starts and ends

Alternative promoters change where an RNA begins. Alternative cleavage and polyadenylation sites change where its processed 3' end is formed. These choices change transcript identity even when the RNA retains the same internal exons.

For example, two RNAs with the same coding region and different 3' UTR lengths encode the same amino-acid sequence. Their different UTRs still distinguish the transcripts.

2.7 Isoforms: a precise distinction among transcript, protein, and modification states

Use **transcript isoform** for an RNA structure and **protein isoform** for a distinct amino-acid sequence associated with a gene. This keeps RNA differences separate from protein-sequence differences.

A **proteoform** describes a specific molecular form of a protein, accounting for sequence differences and chemical modifications. The term is broader than a modification state. For this chapter, distinguish the amino-acid sequence from modifications added to that sequence after translation.

Continue the example. Two transcript isoforms with identical coding sequences encode the same protein sequence. If that protein is phosphorylated, a phosphate group is attached to it. The modified and unmodified proteins are different proteoforms with the same encoded amino-acid sequence. Other changes include adding sugar groups (glycosylation), adding acetyl groups (acetylation), and cutting the protein chain (proteolytic cleavage).

Question	Object to compare
Do the mature RNA structures differ?	Transcript isoforms
Do the encoded amino-acid sequences differ?	Protein isoforms
Do the complete protein molecules differ in sequence or modification?	Proteoforms

These questions describe different levels of the same molecules. Protein sequence variation also contributes to proteoform variation.

A **canonical transcript** is the representative selected by a database's policy. Selection gives applications a default transcript; the remaining transcript records retain their own identities.

2.8 Translation: from coding RNA to protein

Translation makes a chain of amino acids from the instructions in mRNA. The ribosome reads the RNA in three-base units called **codons**. A start codon establishes the reading frame. Subsequent codons specify amino acids until a stop codon ends translation.

The **coding sequence (CDS)** identifies the region used for protein synthesis. The surrounding UTRs lie outside the region translated into the protein. They help regulate how the RNA is used and how long it persists.

A **polypeptide** is a chain of amino acids. Proteins consist of such chains folded and assembled into functional forms. Enzymes catalyze reactions; transport proteins move substances; structural proteins help organize cells. UniProt records describe proteins and their sequences, functions, and supporting evidence.

A predicted coding sequence specifies a possible translation product. Evidence that the corresponding protein is present in a particular cell is a separate experimental observation.

2.9 Coding and noncoding as annotation properties

A protein-coding gene record represents a locus with evidence for a protein-coding product. A noncoding RNA gene record represents a locus whose annotated product is RNA. A **biotype** is the database's classification label for a gene or transcript, such as protein-coding or lncRNA.

Gene-level and transcript-level labels have different scope. A gene classified as protein-coding may also have transcript models that lack a productive coding sequence. Read the transcript's own annotation when selecting a protein product. A predicted early stop and possible RNA degradation also need their own annotation; establishing a functional noncoding RNA requires evidence of its RNA function.

2.10 Major classes of noncoding RNA

The next page separates RNA classes from labels describing genomic position. This avoids treating function, length, and location as interchangeable classifications.

RNA class	Meaning and role
rRNA: ribosomal RNA	Forms the structural and catalytic core of the ribosome
tRNA: transfer RNA	Connects an mRNA codon to its corresponding amino acid during translation
snRNA: small nuclear RNA	Includes RNAs that participate in the spliceosome
snoRNA: small nucleolar RNA	Guides processing and chemical modification of other RNAs, especially rRNA
miRNA: microRNA	A short regulatory RNA processed from a precursor; guides regulation of target RNAs
lncRNA: long noncoding RNA	A long RNA annotated as noncoding; length and coding status define the group, its members have different functions

This table introduces the RNA classes used in this monograph. **Antisense** describes an RNA's position and orientation relative to another gene. **Enhancer-associated** describes its association with an enhancer. An RNA may belong to a class above and also carry one of these positional labels.

For example, a long noncoding RNA that overlaps a gene on the opposite strand is both a lncRNA and antisense to that gene. The two labels answer different questions.

2.11 Long noncoding RNAs in depth

A **long noncoding RNA (lncRNA)** is an RNA classified as noncoding and conventionally longer than about 200 nucleotides. The length cutoff is a laboratory and classification convention. The members have different molecular mechanisms [18].

To understand a particular lncRNA locus, separate three questions: where it is transcribed, what RNA it produces, and what causes any observed biological effect.

A locus is **intergenic** when it lies between the relevant annotated genes. An **antisense** transcript overlaps another gene on the opposite strand. An **intronic** transcript lies within another gene's intron. **Bidirectional** describes transcription starting near another promoter and proceeding in the opposite direction. These descriptions use different spatial relationships and therefore overlap.

There are three distinct explanations to test when changing a lncRNA locus affects a cell. First, the RNA molecule itself may act, for example by binding a protein. Second, transcription or RNA processing at that location may affect a neighboring gene. Third, the DNA region may have a regulatory role independent of the RNA product. More than one explanation may apply to a locus.

Terms such as scaffold, guide, and decoy summarize proposed molecular roles. A mechanistic account must identify what binds what, what changes, and which experiment supports that conclusion.

Perturbation methods help separate those explanations. Suppressing a promoter changes RNA production and activity at the locus. Targeting the RNA molecule more directly tests the contribution of the RNA product. Altering the DNA sequence tests the affected DNA region, with consequences that depend on which sequence was changed.

Record both the intervention method and its target region. The method and target region identify which operation was performed.

2.11.1 lincRNA, lncRNA, and related labels

lincRNA means long intergenic noncoding RNA. It is a subclass of lncRNA defined by genomic position. Every lincRNA is a lncRNA; other lncRNAs occupy overlapping or intronic positions.

Database labels also change between releases. GENCODE's lncRNA category replaced several older, more specific labels [6]. Preserve the exact label from the source release so that later readers know which classification was used.

2.11.2 Expression and evidence

Expression means production of a gene's RNA; an expression measurement estimates its abundance in the sampled material. Detection depends on the cells sampled and the amount of RNA captured. A detection result describes the sampled cells and the sensitivity of the experiment.

Short sequencing reads provide evidence for local RNA sequence and exon junctions. Long reads connect larger parts of an RNA in one observation. Low abundance, repeated sequences, and overlapping genes make it harder to reconstruct a complete transcript. Revised models therefore need identifiable annotation releases.

2.12 cDNA: complementary DNA as an experimental representation of RNA

Complementary DNA (cDNA) is DNA synthesized from an RNA template by the enzyme reverse transcriptase. The enzyme uses base-pairing rules to make a complementary DNA strand. The operation is called **reverse transcription** [20].

Follow one molecule. A mature RNA contains exon-derived segments 1-3-4. Reverse transcription copies that RNA into DNA. The cDNA represents those segments; its template already lacks exon 2 and the introns removed during RNA processing.

A full-length cDNA represents an RNA across its length. A partial cDNA represents only a portion. A **cDNA library** is a collection made from an RNA population. These terms describe the extent of a copy or a collection of copies.

DNA serves as an intermediate for workflows that amplify and sequence DNA. In single-cell RNA sequencing, cell barcodes and unique molecular identifiers are associated with the captured molecules during library preparation. Those labels let analysis connect sequence reads to a cell partition and distinguish original captured molecules from amplified copies [16,17].

2.12.1 cDNA, genomic DNA, and transcript models

These three objects occupy different steps between the cell and the database.

Object	What it is	What it represents
Genomic DNA	The DNA in a chromosome	The locus, including its introns and surrounding sequence
cDNA	A physical DNA copy made from an RNA template	The RNA sequence captured by the experiment
Transcript model	A database description of an inferred RNA structure	The transcript boundaries and exon arrangement supported by evidence

A genome assembly is a sequence representation of genomic DNA. It belongs to the computational description, distinct from the physical chromosome itself.

In our example, genomic DNA contains exons 1, 2, 3, and 4 separated by intervening sequence. The captured mature RNA contains 1-3-4. Its cDNA therefore represents 1-3-4. A transcript model records the corresponding exon coordinates and links the structure to the gene.

A cDNA observation supports the portion of the transcript it covers. Annotation combines such observations with other sequence evidence and curation to infer the complete structure. Multiple matching cDNA molecules strengthen support for the same transcript model.

Terminology note. Laboratory cDNA means a DNA molecule copied from RNA. Coding-DNA coordinates in variant nomenclature describe positions relative to a coding reference transcript. The context determines which meaning is intended.

2.13 Pseudogenes in depth

For the protein-derived pseudogenes discussed here, a **pseudogene** is a genomic sequence descended from a protein-coding gene that has lost the ability to produce the ancestral functional protein. Sequence resemblance and evidence of disrupted coding capacity support its classification [6,19].

For example, a copied gene may acquire a deletion that shifts its reading frame. The sequence still resembles its ancestral gene. The mutation disrupts the instructions needed to produce the ancestral functional protein. This explains both parts of the term's use: gene-like ancestry and loss of the ancestral coding function.

Pseudogene classification describes that ancestry and coding status. Whether the locus is transcribed, and whether its RNA has an effect, are separate questions. They require their own evidence.

The following subsections distinguish three routes of origin. They ask how the sequence arose. Expression describes its activity today.

2.13.1 Duplicated or unprocessed pseudogenes

A **duplicated pseudogene** originates from duplication of genomic DNA. The duplicated region includes the gene sequence and its exon-intron structure. One copy retains the ancestral protein-coding function while another accumulates disabling changes.

For an illustrative case, begin with a gene containing three exons separated by two introns. DNA duplication produces a second copy with the same arrangement. A later disabling mutation in the second copy makes it a pseudogene. Because the copy arose from genomic DNA, its retained introns help identify its origin. This class is also called **unprocessed**.

2.13.2 Processed pseudogenes

A **processed pseudogene** originates through an RNA intermediate. A mature RNA is reverse-transcribed into DNA, and the copy is inserted into the genome. This insertion process is called **retrotransposition**.

If the original mature RNA joined exons 1-2-3 after removing the introns, the inserted DNA contains that joined sequence. Its lack of the original introns is evidence of an RNA-derived origin. A remnant of the RNA's poly(A) tail provides another clue.

The original promoter remains at the source DNA locus. Expression at the new site depends on the surrounding regulatory context. An RNA-derived insertion is classified as a pseudogene when the evidence supports loss or absence of the ancestral functional protein product; some RNA-derived insertions retain protein-coding function.

2.13.3 Unitary pseudogenes

A **unitary pseudogene** arises when an ancestral gene loses its protein-coding function in a lineage while that lineage lacks a functional duplicate.

Consider an ancestral gene inherited by two species. One species retains a functional version. In the other, disabling mutations disrupt the original gene. The disrupted locus is unitary because that lineage lacks a retained functional duplicate. Comparing the two species supplies evidence for the ancestral function.

Route of origin	What happened?	What distinguishes it?
Duplicated / unprocessed	Genomic DNA was copied, then a copy lost function	Origin by DNA duplication
Processed	RNA was copied back into DNA and inserted	Origin through processed RNA
Unitary	The ancestral gene lost function in a lineage	Loss of function at the original locus

These are the three origin categories used in this introductory account. Transcription and translation evidence are recorded separately, as the next subsection explains.

2.13.4 Transcribed and translated pseudogene evidence

A **transcribed pseudogene** has evidence that its locus produces RNA. The evidence must distinguish that locus from related sequences. GENCODE uses labels such as transcribed processed pseudogene and transcribed unprocessed pseudogene [6].

Translation evidence asks a further question: was a peptide produced? Demonstrating restoration of the ancestral functional protein requires evidence of that protein's function. A peptide fragment and a functioning ancestral protein are different observations.

For data integration, retain separate fields for locus classification, RNA evidence, and protein evidence. An RNA measurement for a pseudogene is useful even when a protein-based model input lacks a supported sequence.

2.13.5 Why pseudogene symbols often resemble parent-gene symbols

A parent-derived symbol communicates a pseudogene's relationship to its ancestral gene or gene family. HGNC naming conventions include a P marker in pseudogene symbols, with details depending on the naming class [1,23].

Use the stable HGNC identifier to identify the record and the recorded locus type to establish its classification. Symbol spelling helps a reader recognize a relationship; the database record supplies the authoritative label.

2.13.6 Pseudogenes in expression data

Pseudogene sequences often resemble their parent-gene sequences. That similarity creates a concrete measurement problem. If a short RNA-derived read matches both loci equally well, the read's source is ambiguous.

An analysis must apply its read-assignment rules. Reads with distinguishing sequence provide stronger evidence for a particular locus. Ambiguous reads need the handling specified by the analysis method. The resulting expression value therefore depends on both the annotation and the assignment procedure.

When a measurement table contains a pseudogene column, preserve the column's gene identifier and source annotation. Keep the display symbol and local column key in their own fields, just as for other measured genes.

A useful distinction. Classification asks what sort of locus the database represents. Expression asks how much RNA the experiment assigned to it. Protein evidence asks whether a translated product was observed or otherwise supported. None of these fields substitutes for the other two.

2.14 Paralogs, orthologs, and gene families

Homologous genes share evolutionary ancestry. Two terms distinguish the event that separates them: **paralogs** arise through gene duplication, and **orthologs** diverge through speciation.

For a duplication example, an ancestral gene is copied into genes A and B. A and B are paralogs. For a speciation example, an ancestral species divides into two descendant species, each inheriting a version of gene A. Those two versions are orthologs.

The defining distinction is duplication versus speciation. Later speciation also distributes descendants of paralogs across species. A **gene family** groups related genes descended from a shared ancestral gene; its members include relationships created by duplication and divergence.

Pseudogenes also retain evidence of these ancestral relationships. Their similar sequences and names help reveal family membership. Each member retains its own identity.

For cross-species work, record the organism alongside the gene identifier. A family name identifies a relationship; a species-qualified database identifier identifies the particular gene record being used.

2.15 A compact biological object graph

The chapter's relationships fit into the following sequence. The left column names a biological object or event; the right column names the record used to describe it.

Biology	Representation used in analysis
DNA occupies a region of a chromosome	An interval on a named genome assembly
A gene is transcribed into RNA	A gene record linked to transcript models
RNA processing produces a particular mature structure	An ordered set of exon intervals and transcript boundaries
A coding RNA specifies an amino-acid sequence	A coding sequence linked to a protein sequence or entry
A protein is chemically modified	A description of the resulting proteoform

Regulatory DNA and regulatory molecules influence transcription. Experimental interventions alter a chosen molecule, DNA region, or process. Sequencing observes molecules, and analysis assigns those observations to reference records.

The identifier system preserves those steps by naming each object separately. It records which gene a transcript belongs to, which sequence a coding transcript specifies, which intervention was performed, and which measurements entered the model.

Source note for chapters 1-2. Numbered citations refer to Appendix G of this monograph. These chapters revise the original exposition and examples. MANTRA implementation claims in later chapters retain the original document's evidence status. GENCODE's biotype definitions [6] supply the named annotation categories. The working distinction between encoded sequence and complete protein form follows the proteoform terminology of Smith and Kelleher, *Nature Methods* 10, 186-187 (2013), doi:10.1038/nmeth.2369. The examples with numbered exons are illustrative.

3 Genome Annotation

3.1 Assembly and annotation solve different problems

A **genome assembly** provides reference DNA sequences and a coordinate system. A **genome annotation** identifies features on those sequences: genes, transcripts, exons, and coding regions. The assembly answers, "Which bases occupy this position?" The annotation answers, "Which biological feature occupies this region?"

GRCh38.p14 identifies an assembly. GENCODE release 44 identifies an annotation built on that assembly and paired with Ensembl release 110 [5]. To summarize a chromatin measurement near a gene, an analysis needs both: the assembly locates the measurement, and the annotation supplies the gene's boundaries.

An assembly can remain fixed while its annotation changes. New RNA evidence may reveal an additional exon, so annotators revise the transcript model while keeping the same reference DNA. Conversely, a revised assembly may change a gene's numerical coordinates while retaining the same interpretation of the gene.

3.2 What an annotation team actually constructs

An annotation team turns experimental observations into records describing where genes lie and how their transcripts are organized. The input is a reference assembly plus evidence from RNA sequencing, cDNA, proteins, comparisons between species, and published experiments.

Consider the four-exon example from chapter 2. Reads support one RNA joining exons 1-2-3-4 and another joining 1-3-4. The team represents those observations with two transcript models linked to one gene model. Each transcript model records its own exon coordinates. For a coding transcript, the annotation also marks the portions that specify its protein sequence.

The resulting records have an explicit structure:

Record	What it stores	What it links to
Gene	Locus, identifier, name, and classification	Its transcript records
Transcript	RNA boundaries, exon arrangement, and classification	Its gene, exons, and any coding sequence
Exon	One retained genomic interval	The transcript in which it occurs
Coding sequence	The translated portions of a coding transcript	The transcript and its translated sequence

Annotators attach support information and cross-references to these records. A gene symbol names the gene for readers; the identifiers and parent links let software reconstruct the structure.

3.3 Sources of annotation evidence

Each kind of evidence answers a different question. Short reads reveal local sequence and splice connections. Longer molecules connect those observations into transcript structures. Protein and comparative evidence help interpret coding regions, and curators examine conflicts among the sources.

3.3.1 Short-read RNA sequencing

A **read** is a sequence reported by a sequencing instrument. In short-read RNA sequencing, a read covers a small part of an RNA-derived molecule. A read spanning the end of exon 1 and the start of exon 3 supports the splice connection 1-to-3.

A collection of reads also shows which regions were expressed in the sample. The limitation is the missing connection between distant observations. Separate reads supporting 1-to-2 and 3-to-4 leave open whether both junctions occurred in the same RNA molecule.

3.3.2 Long-read RNA sequencing

A long read covers a larger part of an RNA or cDNA molecule. A read spanning exons 1-2-3-4 directly connects all three junctions in that observed molecule. This helps distinguish complete transcript structures that short fragments leave ambiguous.

Read errors and incomplete sampling still matter. Annotation teams compare long reads with other evidence to distinguish supported structures from sequencing errors and partially observed molecules.

3.3.3 Full-length cDNA and expressed-sequence tags

A **full-length cDNA clone** is a laboratory DNA copy representing an RNA across its length. Sequencing the clone provides evidence for the processed RNA's sequence and exon arrangement.

An **expressed-sequence tag (EST)** is a shorter sequence obtained from cDNA. It supports the part of the transcript that it covers. cDNA clones and ESTs supplied transcript evidence before modern high-throughput RNA sequencing and remain part of the history behind existing annotations.

3.3.4 Protein evidence

An **open reading frame** is a run of codons that could specify a protein before a stop codon is reached. Annotators compare a candidate translated sequence with known proteins and conserved protein regions. Peptides detected by mass spectrometry provide experimental evidence for translated portions.

Protein evidence helps locate and assess the coding sequence. The annotation records the connection to the protein evidence so that a reader can trace the conclusion to its source.

3.3.5 Comparative genomics

Comparative genomics compares related sequences across species. If a region preserves a coding sequence and splice boundaries across related species, that agreement supports the proposed gene structure.

The same comparison helps identify unitary pseudogenes. A functional gene in one species and a disrupted counterpart in another provide evidence that the disrupted gene descended from a functional ancestral gene.

3.3.6 Manual curation and literature

Curators inspect the evidence when candidate models conflict. They compare sequence alignments, experiments, neighboring genes, and record history, then document the supported interpretation. This is especially useful for overlapping or readthrough transcripts, similar gene-family members, pseudogenes, and clinically studied loci. GENCODE combines manual Ensembl-Havana curation with computational annotation and targeted evidence generation [4].

3.4 Gene models and transcript models

A **gene model** represents a locus and its associated transcripts. A **transcript model** represents one inferred RNA structure in a particular annotation release. Its fields fall into three groups:

Group	Information retained
Location and structure	Reference sequence, strand, start and end, ordered exon coordinates, splice junctions, UTRs, and coding coordinates where defined
Identity and classification	Gene and transcript accessions, feature versions, gene and transcript biotypes
Evidence and links	Support attributes, tags, and cross-references

The model describes a class of RNA molecules sharing the represented structure. Individual molecules are experimental observations that support or match that model.

3.5 Why transcript annotations change

A transcript model changes when the evidence changes its interpretation. A newly observed exon changes the structure. A corrected sequence changes the recorded bases. Revised locus boundaries may change the gene to which a transcript belongs. Reconciliation may merge or split records, and coding evidence may change a transcript's classification.

Ensembl tracks continuity with a stable identifier and records qualifying changes with a version suffix [3]. Gene versions track changes to linked transcripts; transcript versions track changes such as splicing, location, or cDNA sequence; protein versions track peptide-sequence changes; exon versions track changes to exon sequence. A record is retired when it fails the stable-ID mapping procedure's continuity rules.

For ENST00000611639.4, the stable stem is ENST00000611639 and the feature version is 4. The annotation release identifies the collection in which that version appears. A feature version and a database release therefore identify different levels of change.

3.6 Canonical, representative, principal, and MANE transcripts

A **representative transcript** is a selected default for a particular use. The database's selection policy determines what labels such as canonical or principal mean. Selection criteria include transcript support, expression, coding conservation, protein evidence, and clinical relevance.

Suppose a gene has three annotated transcripts. An application that needs one default chooses a transcript under a stated policy and records its accession. Another application may choose a different transcript because it needs different coding or clinical coverage. The other transcript records remain available.

MANE matches selected RefSeq transcripts to Ensembl/Gencode transcripts. Each matched pair agrees in sequence, exon coordinates, and coding region. MANE Select supplies one representative transcript per included locus; MANE Plus Clinical adds transcripts needed to represent clinically relevant variants beyond that selection [26].

CCDS records protein-coding regions agreed upon by collaborating annotation groups. Its scope is the consensus coding sequence. A full transcript also includes untranslated regions. Record the resource, selection policy, release, and chosen accession when using either kind of representative.

3.7 GTF and GFF as annotation exchange formats

A **GTF** file stores annotation as tab-separated rows. It belongs to the GFF family of feature-description formats; parsers must follow the particular format's rules. Each GTF row identifies one feature and its genomic interval [25].

Column	Meaning
seqname	Chromosome or other reference sequence
source	Producer of the annotation
feature	Feature type: gene, transcript, exon, CDS, start_codon, or stop_codon
start, end	One-based inclusive interval boundaries
score	Numeric score, or a missing-value marker
strand	Forward (+) or reverse (-) orientation
frame / phase	Coding phase for a CDS feature; 0, 1, 2, or missing
attributes	Identifiers, names, biotypes, tags, and cross-references

The final column connects the rows. A gene row supplies a `gene_id`. A transcript row supplies that `gene_id` and its own `transcript_id`. Exon and CDS rows repeat the relevant parent identifiers.

For example, attributes containing `gene_name` "TP53" and `gene_type` "protein_coding" name and classify the gene. The `gene_id` and `transcript_id` fields establish which records belong together. A parser uses those keys to reconstruct the gene-to-transcript-to-exon relationships.

Record the exact GTF input: release, assembly, source URL, byte count, and SHA-256. A release may offer several files with different subsets. The file identity establishes which records were actually available to the analysis.

3.8 Gene and transcript biotypes

A **biotype** is an annotation category assigned to a gene or transcript. Apply it at the level where it was recorded. A protein-coding gene may include a retained-intron transcript with a different transcript-level classification.

Pseudogene biotypes distinguish origin and expression evidence. Noncoding RNA biotypes classify RNA products and, in some vocabularies, their genomic relationships. Preserve the source's exact label. If the project combines several source labels into one downstream category, keep that translation in a versioned mapping table [6].

3.9 Genome assembly patches and alternate loci

A patch release adds corrections or alternative representations while retaining the established primary-chromosome coordinates. A **fix patch** supplies corrected sequence intended for a future assembly. A **novel patch** supplies an alternative representation of a variable region.

Reference bundles also distinguish primary chromosomes, alternate loci, and unplaced contigs. Some analysis bundles add decoy sequences to absorb reads that would otherwise align misleadingly. A gene's recorded coordinates must identify the component sequence being used.

The reference bundle, alignment index, annotation file, and counting configuration together determine which sequences and features contribute to an expression matrix.

3.10 Coordinate transformations and lift-over

Lift-over translates an interval from one assembly to another using alignments between reference sequences. An interval may map once, map to several candidate locations, or fail to map. Rearrangements and sequence differences may also change its length or split its mapping.

Keep the source interval, destination interval, mapping method, and result status. For gene-level integration, the document favors interpreting each source in its own pinned reference and then joining established gene or transcript records. Use lift-over when the task requires transferring the intervals themselves, as with a signal track.

3.11 Release pinning as scientific state management

Pinning a release means recording and reusing a specific published database state. Pinning its source file also identifies the exact records and bytes used. The MANTRA annotation snapshot reported in this document is listed on the next page.

Field	Reported MANTRA value
GENCODE release	44
Ensembl release	110
Genome assembly	GRCh38.p14
Annotation source	The project's GENCODE comprehensive annotation file
Source-file SHA-256	01f817afed65fee863361b4baf30a95e722ee5c5d508ff77b04106ef7ba20d3

GENCODE's release history confirms the pairing of release 44, Ensembl 110, and GRCh38.p14 [5]. The source-file hash identifies the particular annotation input reported by MANTRA.

These values answer three separate questions. Release 44 selects the GENCODE annotation package. Ensembl 110 identifies the paired genome-database release. GRCh38.p14 supplies the reference sequences and coordinates. The SHA-256 identifies the exact file used from that package.

How to use the snapshot. When interpreting an old transcript identifier, first look it up in the annotation release used by the original analysis. Then, if needed, compare it with a later release. Keep both lookup results so that a reader can distinguish the original interpretation from the updated one.

Project-specific counts, hashes, mappings, and array conventions throughout this monograph refer to the pinned MANTRA materials reported by the source document [24]. Here, "current" means current within the named snapshot. This revision preserves the reported project evidence; verification of a live checkout is a separate task.

4 The Identifier Systems

4.1 The central claim: these nomenclatures are orthogonal

A row about one experiment may contain a gene symbol, transcript accession, protein accession, guide-pair ID, and model key. Each field names a different object. The row connects them because the experiment or analysis needs those relationships.

The term **orthogonal** in this document means separating those questions in the data model. HGNC supplies nomenclature. Ensembl supplies feature records and identifiers. GENCODE supplies an annotation release. RefSeq supplies reference sequences. UniProt supplies protein records. ENCODE supplies experimental records and files. MANTRA supplies local keys for interventions and model coordinates.

The systems share information through cross-references. Follow one protein-coding example:

Step	Record and relationship
Identify the gene	An HGNC record links to an Ensembl gene record
Identify an RNA structure	The Ensembl gene links to its transcript records in a release
Identify a reference RNA	A supported cross-reference connects a transcript to a RefSeq RNA record
Identify a protein	The coding transcript supplies a translated sequence, linked to protein records
Add a measurement	An ENCODE experiment supplies data associated with the gene's region
Use the result in MANTRA	A project key fixes the result's row or column position

Each step retains the identifiers at both ends of the link. A transcript accession identifies the RNA record; a protein accession identifies the protein record reached through that relationship.

For an ENCODE example, a ChIP-seq experiment measures where a selected protein is associated with chromatin. The experiment has its own accession. Its measurements are located on reference DNA and summarized over annotated regions. The experiment accession, gene identifier, and assembly therefore belong in separate fields.

The next table is a lookup guide: begin with the question your analysis needs to answer, then choose the corresponding resource.

4.2 A question-to-resource matrix

Question	Resource or field	Object returned
What is the approved human gene symbol?	HGNC	Nomenclature record and symbol
Which genomic feature is this?	Ensembl ENSG, ENST, ENSP, ENSE	Gene, transcript, protein translation, or exon record
Which structures belong to this annotation?	GENCODE release and GTF	Gene and transcript features in the release
Which NCBI gene record is this?	NCBI GeneID	Gene record linking its products and annotations
Which reference sequence is being used?	RefSeq accession.version	RNA, protein, or genomic sequence record
Which protein entry or isoform is being used?	UniProt accession and isoform	Protein knowledgebase entry or represented isoform
Which experiment or file supplied the data?	ENCODE accession	Experiment, sample, file, library, or pipeline record
Which reference defines these coordinates?	Assembly name or accession	Reference sequence collection
Which intervention row is this?	MANTRA perturbation identity	Preserved split-contract row
Which predicted expression column is this?	resolved_gene_key	Unique model response coordinate

4.3 HGNC: human gene nomenclature

The **HUGO Gene Nomenclature Committee (HGNC)** assigns approved names and symbols to human genes. It also assigns a stable HGNC ID to each nomenclature record. A symbol is the readable label; the HGNC ID identifies the record through naming updates [1,2].

HGNC field	Meaning
HGNC ID	Stable nomenclature-record identifier
Approved symbol	Current short label in the snapshot
Approved name	Full gene name
Previous symbols	Earlier approved labels recorded for the gene
Alias symbols	Other labels associated with the gene
Locus type	Recorded gene or locus classification
Chromosomal location	Summary of the gene's location

HGNC field	Meaning
Gene group	Curated family or functional grouping
Cross-references	Links to Ensembl, NCBI Gene, RefSeq, UniProt, CCDS, and other records

A nomenclature resolver searches a pinned HGNC snapshot and records how the input matched: approved symbol, previous symbol, alias, or another stated relation. Keeping that match type explains why a historical experimental label resolves to a differently named record.

4.3.1 Approved symbols, previous symbols, and aliases

An **approved symbol** is the official symbol in the HGNC snapshot. A **previous symbol** records an earlier approved label. An **alias** is another name associated through literature or database usage. These are separate fields because they describe different naming histories.

If an alias matches two records, the lookup has two candidates. A resolver must retain the candidate set and report ambiguity until the available evidence identifies one record. A name match requires additional evidence to establish a unique biological identity [2].

4.3.2 HGNC locus types

HGNC records cover protein-coding genes, noncoding RNA genes, pseudogenes, immunoglobulin and T-cell-receptor genes, readthroughs, regions, and other recorded features. The **locus type** names the classification of the HGNC record.

Use that classification when assessing whether a protein-based input is appropriate. Then consult transcript and protein evidence to select the actual sequence. The gene-level label and the transcript-level evidence have different scopes.

4.3.3 Example: ALG1L and ALG1L1P

The reported atlas uses ALG1L as an experimental perturbation label. The pinned HGNC snapshot records ALG1L as a previous symbol for ALG1L1P, a pseudogene with HGNC ID HGNC:33721 [24].

Retain ALG1L as the experimental row label. Store ALG1L1P and HGNC:33721 as the nomenclature interpretation. The original row and the resolved record remain connected and separately identifiable.

4.4 Ensembl: stable identifiers for genome features

Ensembl is a genome database that assigns identifiers to annotated features. Human prefixes distinguish these feature types [3]:

Prefix	Object	Example
ENSG	Gene feature	ENSG00000189366
ENST	Transcript feature	ENST00000611639
ENSP	Translated protein feature	ENSP00000482705
ENSE	Exon feature	ENSE00003979176

A stable stem and a version suffix identify different parts of a record's history. In ENSG00000189366.13, the stem is ENSG00000189366 and the gene-feature version is 13. ENST00000611639.4 similarly identifies version 4 of a transcript feature.

Ensembl compares features across releases and retains stable IDs when its continuity rules are satisfied. Qualifying feature changes increment the version [3].

4.4.1 Ensembl gene, transcript, and protein relations

To find a protein associated with an Ensembl gene, look up that gene's transcripts in a specified release, then follow the coding transcript's translation link. Each link comes from the annotation.

The lookup requires an annotation relationship between the gene and its translated products. Gene, transcript, and protein records have separately assigned accessions. Release changes may also revise transcript membership or retire a feature, which is why the lookup must name its release.

4.4.2 Species context

Human identifiers use prefixes such as ENSG, ENST, and ENSP. Mouse identifiers include ENSMUSG, ENSMUST, and ENSMUSP. A parser uses the prefix as a type and species hint, while the record retains an explicit organism or taxonomy field.

4.5 GENCODE: a release-specific human and mouse gene annotation

GENCODE publishes gene and transcript annotations for human and mouse. Its evidence combines experiments, manual curation, computational analysis, and targeted investigation [4]. The annotation is distributed in release files and displayed through resources including Ensembl and the UCSC Genome Browser.

GENCODE uses Ensembl gene and transcript identifiers. The GENCODE release specifies the package of structures, coordinates, biotypes, and tags in which those identifiers are interpreted.

Layer	Example	What it identifies
Gene accession	ENSG00000189366	Stable gene-feature record
Feature version	.13	Revision of that feature
GENCODE release	44	Published annotation package
Ensembl release	110	Paired genome-database release
Assembly	GRCh38.p14	Reference sequences and coordinates
GTF SHA-256	Hash of the selected file	Exact annotation input bytes

4.5.1 Comprehensive and basic annotations

GENCODE distributes several files per release. The **comprehensive** annotation contains the full set for its stated scope. The **basic** annotation selects a representative transcript subset under GENCODE's policy. Other distributions restrict the reference components or provide specialized features and metadata [22].

Record the filename and hash as well as the release. Two files from release 44 may contain different transcript sets because they serve different purposes.

4.5.2 GENCODE tags and transcript support

A **tag** records a property such as representative status, cross-resource agreement, or a special annotation condition. Support attributes describe the evidence behind a transcript model.

An analysis that selects transcripts by tag should retain the selected accessions, source records, and selection rule. This lets another reader reproduce both the input annotation and the subset chosen from it.

4.6 NCBI Gene: stable gene records in the NCBI ecosystem

NCBI Gene assigns a numeric GeneID to a gene record; Entrez Gene ID is the historical name for the same system. The record gathers names, type, genomic locations, RefSeq products, orthology, phenotype and literature links, and cross-references. Human nomenclature comes from HGNC where available [9].

The ALG1L1P example in the pinned materials has GeneID 200810. That number names the NCBI gene record. Any linked RefSeq RNA or protein has its own accession. NCBI also records replaced or discontinued gene records. Preserve the source export or snapshot when reproducing a mapping.

4.7 RefSeq: accessioned reference sequences

RefSeq publishes reference sequences for genomes, transcripts, and proteins. Its accessions identify sequence records. Prefixes distinguish sequence classes and model status [10].

Prefix	Sequence class
NM_ / NR_ / NP_	Reference protein-coding RNA / noncoding RNA / protein
XM_ / XR_ / XP_	Computational model RNA / noncoding RNA / protein
NG_	Genomic region, including RefSeqGene records
NC_	Reference chromosome or complete genomic molecule

These prefixes guide interpretation. Curation and evidence details belong to the individual record; experimental support must be assessed from the record's evidence.

In NM_000546.6, the suffix .6 identifies sequence version 6. A sequence change increments that version; a metadata-only update may leave it unchanged. Cite accession and version together when specifying an exact RefSeq sequence [10,21].

RefSeq and Ensembl/GENCODE maintain separate transcript records. MANE explicitly identifies matched pairs. Other cross-references require inspection of their relation: one record may link to several candidates, or lack a corresponding record in the selected release.

4.8 UniProt: protein knowledge and sequence identity

UniProtKB records proteins, their sequences, and supporting knowledge. Entries describe function, domains, catalytic activity, location, modifications, interactions, disease associations, and evidence. Swiss-Prot entries are reviewed and manually annotated; TrEMBL entries are unreviewed and computationally annotated [7,8].

A primary accession such as P49356 identifies an entry. An isoform identifier such as P04150-2 selects a represented protein isoform. Here, -2 is an isoform suffix. Compare ENST00000611639.4, where .4 is a transcript-feature version.

A reviewed entry may organize several isoforms under one accession. Select the sequence through the gene-to-transcript-to-protein evidence and the project's isoform policy. A locus classified as a pseudogene may yield an explicitly empty set of supported protein accessions under that policy.

For a sequence-derived model input, retain the accession, isoform selection, exact sequence bytes, release or retrieval date, source location, and hash. The selected sequence is what the embedding or similarity computation actually reads.

4.9 ENCODE: identifiers for functional-genomics data

ENCODE publishes experimental data and the records needed to interpret them. The accession prefix identifies the record type [11].

Prefix	Object	What it identifies
ENCSR	Experiment	One assay experiment
ENCFF	File	One raw or processed data file
ENCBS	Biosample	A prepared biological sample
ENCLB	Library	A sequencing library
ENCDO	Donor	A human donor or model-organism strain record
ENCAB	Antibody lot	A characterized antibody reagent
ENCPL	Pipeline	A data-processing pipeline record

An ENCODE signal track supplies measurements along reference DNA. A GENCODE annotation supplies the intervals used to summarize those measurements. Their assembly identifies the common coordinate system. Store the file accession, annotation release, and assembly separately so that the computation is traceable to all three inputs.

4.10 Genome Reference Consortium and assembly identifiers

GRCh37 and GRCh38 are human assembly names from the Genome Reference Consortium lineage. Assembly accessions identify the sequence collection in sequence archives. Component accessions identify individual sequences; NC_000001.11, for example, is a RefSeq chromosome-sequence accession.

The MANTRA coordinate-derived resources described here use GRCh38.p14 together with the specified GENCODE input. For an external source on another assembly, record its original assembly and any coordinate transformation used to integrate it.

4.11 MANE and CCDS as convergence layers

A MANE relation connects a RefSeq transcript with a matching Ensembl/GENCODE transcript. The two records retain their accessions while the relation states their agreement in a named release [26].

A CCDS ID identifies an agreed coding-sequence record. Keep it beside the full-transcript identifiers because the coding region is one part of the transcript. MANE and CCDS make particular agreements explicit across resources.

4.12 A formal crosswalk model

A crosswalk is a set of recorded pairs. For example, each HGNC-to-Ensembl pair connects one HGNC record to one Ensembl gene feature. One source record may occur in several pairs.

Let $G_H^{(h)}$ denote HGNC gene records in snapshot h ; $G_E^{(r)}$ and $T_E^{(r)}$ denote Ensembl gene and transcript features in release r ; $S_R^{(s)}$ denotes RefSeq sequence records in snapshot s ; $P_U^{(u)}$ denotes UniProt entries in release u . The notation $A \times B$ means all possible ordered pairs drawn from sets A and B . A relation M is the subset actually supported by the source.

$$M_{HE}^{(h,r)} \subseteq G_H^{(h)} \times G_E^{(r)}$$

$$M_{ET}^{(r)} \subseteq G_E^{(r)} \times T_E^{(r)}$$

$$M_{TR}^{(r,s)} \subseteq T_E^{(r)} \times S_R^{(s)}$$

$$M_{TP}^{(r,u)} \subseteq T_E^{(r)} \times P_U^{(u)}$$

These relations respectively connect nomenclature to gene, gene to transcript, transcript to RefSeq sequence, and transcript to UniProt entry. The superscripts name the source snapshots. A stored crosswalk table must retain those snapshots and the matching procedure that supplied its rows.

4.13 Identifier syntax reference

Identifier	Namespace and object	Version or context
HGNC:33721	HGNC gene nomenclature record	Snapshot-level context
ENSG00000189366.13	Ensembl gene feature	.13 feature version
ENST00000611639.4	Ensembl transcript feature	.4 feature version
ENSP00000482705.1	Ensembl protein feature	.1 feature version
200810	NCBI gene record	Snapshot
NM_000546.6	RefSeq RNA sequence	.6 sequence version
NP_000537.3	RefSeq protein sequence	.3 sequence version
P04637	UniProt protein entry	Entry release
P04150-2	UniProt protein isoform	-2 isoform suffix
ENCSR868FGK	ENCODE experiment	Portal record state
ENCFF948AFM	ENCODE file	Record and downloaded bytes
GRCh38.p14	GRC assembly	Patch release 14
ALG1L	MANTRA intervention row	Split contract and source atlas
TBCE-1	MANTRA response key	Response-axis contract

4.14 Orthogonality in one concrete row

This reported ALG1L example preserves the experimental label while adding its interpretation under the pinned resources [24]:

```
perturbation_identity: ALG1L
match_kind: previous_symbol
hgnc_id: HGNC:33721
approved_symbol: ALG1L1P
ensembl_gene_id: ENSG00000189366
raw_transcript_label: ENST00000340333.3
current_gencode_transcript: ENST00000611639.4
ncbi_gene_id: "200810"
locus_type: pseudogene
uniprot_accessions: []
gencode_release: "44"
ensembl_release: "110"
genome_assembly: GRCh38.p14
```

The raw transcript records the library's design context. The current transcript records the release-44 interpretation. The empty protein list records completed resolution with an empty set of supported accessions under the stated policy. Each field answers a separate question about the same case.

5 Perturbation Experiments and Guide Architecture

5.1 Perturbation as an experimental object

A **perturbation** is an intervention used to change a biological system and measure the response. Interventions include genetic, chemical, environmental, and physical treatments. This chapter covers CRISPR-based interventions.

A target gene name identifies the target; the method identifies the operation: exon cutting, promoter repression or activation, RNA degradation, base editing, or interval deletion. Their consequences depend on the operation and the affected region.

Record the target label, guide or guide pair, construct design, effector system, and library. These fields identify the delivered intervention and explain how it was intended to act.

5.2 CRISPR-Cas targeting from first principles

In the DNA-targeting systems used here, a **guide RNA** directs a Cas protein to a selected sequence. A **single-guide RNA (sgRNA)** contains a variable targeting sequence and a scaffold that interacts with the Cas protein.

The guide's targeting sequence pairs with complementary DNA. The Cas protein also recognizes an adjacent DNA motif called the **PAM**, or protospacer-adjacent motif. Together, sequence pairing and PAM compatibility determine candidate target sites. The attached effector determines the operation performed there.

Effector	Intended operation at the target
Active Cas9 nuclease	Cut both strands of DNA
dCas9	Bind DNA with cutting activity disabled
dCas9 with a repressor	Reduce transcription: CRISPRi
dCas9 with an activator	Increase transcription: CRISPRa
Base or prime editor	Make the sequence changes supported by that editor

Assign a guide to a gene using its sequence, reference assembly, target region, and library design. The assignment is a recorded relationship between the guide and the intended biological target.

5.3 CRISPR nuclease knockout

A Cas9 knockout design directs an active nuclease to a DNA site, often in a coding exon. Repair introduces a distribution of sequence outcomes. An insertion or deletion that disrupts the reading frame is one route to loss of gene function.

A gene-level knockout label records the intent. Actual disruption depends on guide activity, repair, the cell's gene-copy number, and which functional copies were altered. Preserve the target site and guide design so that the intended intervention and observed outcome remain distinguishable.

5.4 CRISPR interference

CRISPR interference (CRISPRi) uses dCas9 with a transcriptional repressor, such as a KRAB-containing effector. The guide directs the complex to a promoter or transcription-start region. The effector recruits repressive machinery and reduces RNA production. This reduction is called **knockdown**.

Promoter choice matters. If a gene uses two promoters, guides near one promoter act in that promoter's context. A library's transcript label may therefore document the start site used for guide design. Preserve that historical label even when a later annotation reorganizes the transcripts.

The Replogle genome-scale Perturb-seq work used CRISPRi across millions of human cells [13]. Related library work paired guides targeting the same gene to improve repression while keeping a compact set of constructs [14,15].

5.5 CRISPR activation

CRISPR activation (CRISPRa) recruits transcriptional activators through dCas9 to increase expression. Guides select promoter or regulatory sites; the activator changes transcription at those sites.

A pair of guides may address two sites near a promoter or different promoter contexts. Record the sites and design purpose alongside the gene-level label so that the activation experiment is fully specified.

5.6 The semantic layers of a guide record

A guide record separates sequence, location, design interpretation, and delivery position.

Field	Meaning
guide ID	Identifier assigned to the sgRNA in its library
protospacer sequence	Variable sequence used to direct targeting
scaffold / constant-region design	RNA architecture and capture compatibility
target assembly	Reference used for guide design
target coordinate and strand	Intended binding site and orientation
PAM	Adjacent DNA motif required by the Cas system
target gene or transcript label	Library's interpretation of the target
predicted activity	Estimated guide efficacy
off-target metrics	Estimates or candidate sites for unintended targeting
vector position	Position A or B in a paired cassette
construct ID	Identifier of the complete delivered library element

Two guides targeting one gene have different guide IDs. One guide used in two pairs belongs to two construct records. Position A or B also matters when the vector uses different promoters, constant regions, or capture sequences at its two positions.

5.7 Single-guide library elements

A **library element** is one designed construct in a collection of perturbations. A single-guide element encodes one sgRNA. A vector delivers the element to cells, and the experiment measures the cells associated with that guide.

A pooled library may include several independent guides for one gene. Agreement among their effects strengthens the interpretation of a gene-level response. More guides also increase library size and the number of cells needed to represent each construct adequately.

In a single-cell experiment, analysis assigns the guide to a cell using direct guide capture, a linked barcode, or another stated indexing method. The cell record retains the resulting assignment.

5.8 Dual-guide elements targeting one gene

A **dual-guide element** encodes two sgRNAs in one construct. In the same-gene CRISPRi design, guide A and guide B target two sites associated with the same gene's promoter or regulatory region. The delivered pair is one experimental element.

For a concrete comparison, consider two designs aimed at reducing gene A's transcription:

Design	Delivered construct	Intended gene-level intervention
Single-guide	One guide targeting a site near gene A	Repress gene A
Same-gene dual-guide	Two guides targeting distinct sites near gene A	Repress gene A using the pair

The number of guides and the number of targeted genes answer different questions. A same-gene pair contains two targeting sequences and represents one intended gene-level intervention. A combinatorial pair targets two genes, as described in §5.9.

The Replogle dual-sgRNA CRISPRi design selected pairs of high-performing guides and expressed them from a tandem cassette. Direct-capture workflows recovered both guide positions at single-cell resolution [14,15].

Keep both components in the construct record. Gene-level analysis groups the pair under its target gene; construct-level analysis identifies the particular A/B pair delivered.

5.8.1 Why two guides can improve knockdown

A guide must be expressed, bind an accessible target, and place the repressor where it affects transcription. Guide sequence, distance from the start site, chromatin, and effector occupancy influence those steps.

A pair supplies two selected targeting sites. In the reported dual-guide design, their combined action improved knockdown [14]. This is an experimental result for the tested dual-guide design; each new pair still requires an assessment of its own activity.

The measured response belongs to the delivered construct. Preserve the individual guide scores as design metadata and the pair's response as the experimental observation.

5.8.2 Guide A and guide B as positional identities

A and B identify positions in the vector cassette. Swapping positions may change expression or capture because the surrounding sequences and promoters can differ.

Represent an ordered pair as $q = (q_A, q_B, v, c)$: q_A is the guide in position A, q_B the guide in position B, v the vector design, and c the construct identifier. This representation retains both targeting sequences and their delivery context.

5.9 Paired guides targeting different genes

A **combinatorial pair** targets two different genes or loci. A construct targeting genes A and B is a joint intervention. Compare its response with corresponding single-gene interventions to study genetic interactions.

If a design selects two guides for A and three for B, pairing each A guide with each B guide yields six constructs. Their measurements support the gene-pair analysis while retaining guide-level variation. A targeting guide paired with a non-targeting control estimates a single-gene effect within the same two-position architecture.

The name sgID_AB identifies the paired format. The target metadata establish whether the pair represents one gene, two genes, or a deletion design.

5.10 Paired guides for genomic deletions

A flanking deletion design uses two nuclease guides at the boundaries of an interval. Cutting and repair may remove the DNA between the sites. The intended object is that interval, which might contain an exon or a regulatory element.

Record both boundary sites, the effector, and the intended interval. The next table separates flanking deletion from same-gene repression and combinatorial targeting.

Pair architecture	Guide targets	Intended intervention
Same-gene dual CRISPRi	Two sites for one gene or promoter	Repress one gene
Combinatorial pair	Two different genes or loci	Apply a joint intervention
Flanking deletion pair	Opposite boundaries of an interval	Delete the intervening DNA

5.11 The sgID_AB field in the K562 atlas

In the reported atlas, obs/sgID_AB stores the paired-guide identity called for each cell. The ALG1L example displays two components separated by a vertical bar [24]:

```
ALG1L_- _125655482.23-ENST00000340333.3 |
ALG1L_- _125655452.23-ENST00000340333.3
```

Both components retain the perturbation symbol and design-era transcript label. Their coordinate-like portions distinguish the guide sites. Preserve the exact paired string; interpreting its internal syntax requires the library's format definition.

obs/gene and obs/perturbation_id identify the gene-level perturbation group. obs/sgID_AB identifies the called construct within that group. The transcript label preserves the library's transcript or promoter context.

5.12 Direct guide capture

Direct guide capture sequences guide-derived molecules alongside the cell's RNA-expression measurements. The guide design and capture chemistry allow those molecules to carry the cell-barcode information needed for assignment [15].

The experiment produces two measurement streams. RNA-derived reads support gene-expression measurements. Guide-derived reads support guide assignments. The cell barcode connects the streams so that the response and called intervention belong to the same cell record.

A guide call is an analysis of observed guide reads. Retain the calling procedure and confidence, especially when a cell has several guide signals.

5.13 Linked barcodes and direct capture

A **linked-barcode** design assigns a separate expressed barcode to a guide. Sequencing recovers the barcode, and a lookup table maps it to the guide. Direct capture recovers guide-derived sequence itself.

Both methods establish an intervention-to-cell assignment. Their recorded molecules and possible assignment errors differ. The data contract should identify the method, guide field, handling of multiple calls, and representation of uncertain assignments.

5.14 Lentiviral delivery and recombination context

Lentiviral vectors deliver guide constructs to cells. A lentiviral particle packages two RNA genomes. During reverse transcription, template switching between them may create an A/B pair different from the planned library construct.

The paired-guide work described here sequenced both positions and handled recombination explicitly [14]. Store the observed A and B guide calls, the planned construct they are compared with, and the resulting validity or recombination flag. The design table and observed pair answer different questions.

5.15 Multiplicity of infection, singlets, and multiple guide assignments

Multiplicity of infection (MOI) expresses the dose of infectious or transducing units relative to the number of cells. In lentiviral screens, lower doses are used to favor one delivered library element among infected cells. Actual integrations vary among cells; the MOI and an individual cell's integration count are different quantities [27].

A **singlet** is a partition containing one cell. A **multiplet** contains more than one. The following observations require different interpretations:

Observation or cause	Why several guide signals appear
Designed dual-guide cassette	One element encodes two guides
Multiple integrations	A cell receives more than one element
Recombination	The observed A/B combination differs from the planned pair
Ambient guide molecules	Guide signal enters a partition from outside its cell
Cell multiplet	More than one cell contributes molecules to a partition

Guide-calling software uses counts, expected construct architecture, thresholds, and cell metadata to assess these possibilities. Preserve the called construct, component guides, confidence, and multiplicity flags.

5.16 Target identity versus response identity

A pair targeting FNTB identifies an intervention. The cell's expression vector measures RNA associated with thousands of response genes. The target belongs to intervention metadata; each measured gene belongs to the response axis. MANTRA preserves these as separately ordered axes.

5.17 Controls

A **non-targeting control** uses guides designed to lack a relevant target under the library's criteria. It measures background effects of delivery, guide expression, effector, selection, and assay handling. Positive controls target genes with expected phenotypes. Targeting-plus-control pairs assess one side of a paired design.

The literal label non-targeting identifies an experimental control. Store its control type explicitly when it appears in a field otherwise used for gene labels.

5.18 Perturbation identity hierarchy

Choose the identity that matches the question being asked. The atlas retains the following levels:

Level	Question	Record used
Target concept	Which gene or region was intended to change?	Target annotation
Perturbation label	Which gene-level benchmark group contains the cells?	Source symbol in the split contract
Library element	Which designed construct was delivered or called?	Construct or library-element record
Guide pair	Which guides occupy positions A and B?	Exact sgID_AB and component calls
Guide sequence and site	Which sequence directs binding, and where?	Guide sequence, assembly, coordinate, strand, and PAM

In the same-gene dual-guide atlas, the library element and guide pair describe the same construct at different levels of detail. A gene-level group may contain several constructs. Keep their links explicit so that aggregation retains its experimental basis.

The raw transcript field adds design context. The source Ensembl gene ID links the intended target to a locus record. Neither field substitutes for the observed guide pair or the benchmark row label.

For analysis, start at the required level. Compare gene-level responses by perturbation label. Diagnose construct performance by exact pair. Investigate targeting by guide sequence and site. The original cell-to-construct assignments connect all three analyses.

6 Single-Cell Perturbation Data

6.1 What Perturb-seq measures

Perturb-seq combines pooled CRISPR interventions with single-cell RNA sequencing. For each cell, it records a called intervention and expression measurements across many genes. Its basic record is (c_i, q_i, x_i) : c_i identifies cell i , q_i its called guide construct or perturbation, and x_i its vector of gene-expression values.

Each position in x_i corresponds to a particular measured gene. Keeping the individual cell records supports analysis of variation, cell states, trajectories, and distributions. Grouping cells by perturbation produces aggregate response profiles.

The workflow connects intervention to measurement:

Stage	Result
Deliver a guide construct	A cell receives an experimental element
Apply the CRISPR effector	The intervention alters the targeted process to the extent achieved
Capture RNA and guide molecules	The library retains cell-barcode information
Sequence and assign reads	Analysis obtains expression values and guide calls
Join by cell identity	Each expression row is associated with its called intervention

6.2 From cellular RNA to sequencing reads

In the droplet workflow described here, cells are partitioned with barcoded beads and reagents. Oligo-dT primers bind the poly(A) tails of messenger RNA. Reverse transcriptase makes cDNA, and library preparation associates the captured molecules with barcodes and sequencing adapters [16,17].

The prepared library carries several kinds of information:

Component	Purpose
Sample index	Identifies a sequencing sample or library pool
Cell barcode	Identifies a droplet or cell partition within its library
UMI	Tags an original captured molecule for counting
Primers and adapters	Enable amplification and sequencing
cDNA insert	Contains RNA-derived sequence used for assignment

The chemistry determines where these components occur and which read retrieves them. In 10x 3' gene-expression chemistry, molecules from one gel-bead partition share a cell barcode; UMIs support molecule counting within that context.

6.3 Cell barcodes

A **cell barcode** is a sequence selected from the chemistry's allowed barcode set. Molecules captured in one partition receive the same barcode. Analysis groups reads by corrected, accepted barcodes and identifies the partitions treated as cells.

The barcode's scope is its library or run. To combine two libraries, retain a composite key containing both library identity and barcode. Otherwise, identical barcode strings from separate libraries could be mistaken for the same cell.

6.4 Unique molecular identifiers

A **unique molecular identifier (UMI)** is a short sequence associated with a captured molecule before amplification. PCR produces many copies of a cDNA. Reads assigned to the same cell and gene with the same UMI are grouped under the pipeline's counting rules, reducing repeated counting of amplified copies.

For example, ten reads from one amplified cDNA support one captured molecule under the counting model. UMI collisions and sequencing errors complicate that estimate. The counting method specifies how it corrects errors and groups UMI observations.

6.5 Read alignment and gene assignment

The RNA-derived sequence is compared with a reference genome or transcriptome. **Alignment** places a read on a reference; **pseudoalignment** identifies compatible reference targets using sequence matches. Annotation then connects compatible sequence to gene features.

The configuration determines how exon, splice-junction, intron, and multi-locus matches contribute to counts. A Cell Ranger reference, for example, combines a FASTA sequence reference with a GTF annotation. The resulting gene IDs and display names come from that reference bundle.

6.6 Counts, normalization, and transformed expression

Raw UMI counts are nonnegative integers estimating captured molecules. **Normalization** rescales a cell's values to account for its total count. CP10K means counts per 10,000: each gene's count is divided by the cell's count total and multiplied by 10,000. **log1p** then applies the natural logarithm to one plus the scaled value.

The reported MANTRA slim atlas stores log1p CP10K in X [24]. Its name "raw atlas" identifies the source file in this pipeline; X contains transformed expression values. Other analyses may select variable genes or construct residuals after these transformations.

Keep the transformation and axis identities separately recorded. Two matrices with identical dimensions and gene labels may contain different quantities.

6.7 Pseudobulk aggregation

A **pseudobulk profile** combines measurements from a group of cells, such as cells assigned to one perturbation. The grouping determines which cells contribute; the statistic determines what the result means.

For raw counts, an analysis may sum counts and then normalize the total. For transformed values, it may average the transformed measurements or apply a baseline procedure. Those operations yield different quantities, so record the actual procedure.

Preserve the cell-selection rules: quality-control mask, guide assignments, matched controls, cell states, and batch handling. The perturbation identity names the group; response-gene identifiers name the resulting vector positions.

6.8 AnnData and H5AD

AnnData stores a matrix together with its row and column descriptions. **H5AD** is its HDF5-based file format. In this atlas, rows are cells and columns are measured gene features [12].

Component	Information stored
X	Cell-by-gene measurement matrix
obs	Cell metadata: perturbation, guides, QC, batch, and control status
var	Gene metadata: identifiers, display names, and selection flags
layers	Additional matrices with the same cell and gene axes
obsm / varm	Additional arrays aligned to cells / genes
obsp / varp	Pairwise arrays relating cells / genes
uns	Other metadata, parameters, and provenance

The file path identifies storage; the manifest defines meaning. For example, obs/gene is a field name whose interpretation comes from the producing dataset. The contract must state its type, meaning, distinct-value count, and any uniqueness requirement.

6.9 The two dimensions of an expression matrix

One expression row belongs to a cell. One expression column belongs to a measured gene feature. The cell's perturbation label is row metadata.

A cell targeted at gene A still has measurements for genes A, B, C, and the other measured genes. The perturbation label answers what was done to the cell; the column identifier answers which gene's expression a value describes. These roles require separate keys even when the same gene name appears in both.

6.10 The raw K562 atlas as an annotated matrix

The pinned atlas reported in the source document has 310,385 cell rows and 8,563 gene columns, written as shape [310385, 8563]. Its SHA-256 is [24]:

```
51d6c7ecb349d34ec4c0e1bfb44a6d832e4113b26f46a319fc7993c028a5f060
```

Required cell metadata describe perturbation, control status, QC masks, counts, and guides. Required gene metadata describe the atlas column key, symbol, Ensembl gene ID, and project selection flags.

The manifest states that X contains log1p CP10K and that qc_pass_minimal is the default cell-selection mask. It also defines var/in_gwps and related flags. The file hash identifies the bytes; these declarations explain how a consumer should read them.

6.11 Direct guide capture as a second feature space

Direct capture produces guide measurements alongside expression measurements. Guide reads are matched to a guide reference. RNA-derived reads are matched to the expression reference. Their feature IDs identify different objects, and the cell barcode joins the measurements.

A container may retain both measurement matrices. Alternatively, preprocessing stores the resulting guide calls in obs. In either case, retain the connection from the call to the guide observations and calling procedure.

6.12 Controls and matched-control baselines

Non-targeting cells provide observed control expression. MANTRA also uses matched-control procedures to estimate a baseline for treated cells. A residual response describes expression relative to the chosen baseline.

The matching procedure determines which controls contribute and how their values are combined. Record those control identities and the target transformation. Both the treated measurements and the baseline use the same response-gene axis so that corresponding values describe the same genes.

6.13 Data provenance from molecule to model

Trace each modeled value through the transformations that produced it: cell and guide construct; captured RNA; barcoded cDNA; sequencing reads; assigned gene and guide features; molecule counts; normalized or transformed expression; and selected or aggregated model inputs.

Each step needs its inputs and procedure recorded. The resulting chain explains both what a model value means and which experimental observations support it.

7 MANTRA Identifier Architecture

7.1 Architectural goal

MANTRA predicts how gene expression changes after an intervention. Its identifier layer connects historical experimental labels to biological records while keeping model rows and columns in a fixed order.

The model has two axes. The **perturbation axis** identifies interventions using the exact labels and order in the fit, tune, and hold split contract. The **response axis** identifies measured genes using 5,000 unique resolved_gene_key values in GEARS order.

Source branch	Operation	Model-facing result
Cell metadata in obs	Group cells by the original perturbation label; attach biological interpretation	Ordered intervention rows
Gene metadata in var and the GEARS gene table	Match Ensembl gene IDs; retain the atlas column; assign unique response keys	Ordered response columns

Biological cross-references connect the branches when needed. Each branch retains its own order and meaning. An intervention row says what was done; a response column says which gene's expression is predicted.

7.2 Canonical object sets and notation

The following symbols name sets used in this chapter. Their counts are those reported for the pinned project materials [24].

Symbol	Set	Count
C	Atlas cell indices	310,385
P	Raw perturbation labels, including control	2,058
P_{gene}	Gene-target perturbations	2,057
V	Raw atlas expression columns	8,563
Y	GEARS response coordinates	5,000
E	Gene-ID strings recorded by the atlas	Source-defined
T	Transcript IDs or library transcript labels	Source-defined
Q	Paired-guide identities	Source-defined

These sets separate cells, interventions, and measurements before any mapping occurs. A gene-ID string in source metadata must still be interpreted in its field's role.

For each cell i in C , retain $R(i) = (p_i, e_i, t_i, q_i)$, drawn from $P \times E \times T \times Q$. The four entries are its perturbation label, source gene-ID string, transcript or library label, and paired-guide identity.

The perturbation label selects the benchmark group. The gene ID records the intended locus. The transcript label preserves library-design context. The guide pair identifies the called construct. The non-targeting control uses the source's literal control labels, so E and T therefore include both biological accessions and the source's experimental-control strings.

7.3 Raw cell-level fields

The atlas reports the following fields and distinct-value counts [24].

H5AD field	Meaning	Source count
obs/gene	Raw perturbation label	2,058 including control
obs/perturbation_id	Perturbation label	2,058 including control
obs/gene_id	Target gene ID or control label	2,058 including control
obs/transcript	Transcript ID or library label	58 distinct values
obs/gene_transcript	Composite library label	2,273 distinct values
obs/sgID_AB	Exact paired-guide identity	2,273 distinct values

For ALG1L, the reported gene fields contain ALG1L, the source gene ID is ENSG00000189366, and the transcript label is ENST00000340333.3. The composite label is:

```
267_ALG1L_ENST00000340333.3_ENSG00000189366
```

It combines a numeric library prefix, symbol, transcript label, and gene ID. The sgID_AB example contains the two guide components shown in §5.11.

The control value is the literal string non-targeting. Represent it with an experimental-control type or flag. The 2,273 construct-level values and 2,058 perturbation labels describe different groupings: several constructs may belong to one gene-level perturbation group.

7.4 Perturbation split contract

The canonical GEARS split divides the 2,057 gene-target perturbations into ordered groups:

Split	Rows	Row key
fit	1,427	Exact source perturbation symbol
tune	357	Exact source perturbation symbol
hold	273	Exact source perturbation symbol

Labels include FNTB, RBM14, and ALG1L. The project defines `perturbation_identity` as the exact symbol string in `split_contract.json`. The order determines which intervention each row of an aligned array represents.

Model outputs, target arrays, biological inputs, and evaluation files must use that same row order. Store resolved HGNC, Ensembl, NCBI, transcript, and protein identities beside the source key.

7.5 Why the perturbation label remains historical

The original library label connects the intervention to its cells, guides, and stored arrays. A later nomenclature update adds a new interpretation of that intervention; the recorded experiment retains its original key.

For ALG1L, replacing the benchmark key with ALG1L1P would change the string used to join the existing arrays. Keeping ALG1L and adding the HGNC mapping preserves both uses: exact access to the experiment and interpretation under the newer snapshot.

7.6 Expression-gene metadata

The raw expression columns have their own keys and labels:

H5AD field	Reported distinct values	Use
<code>var/_index</code>	8,563 unique keys	Access the exact atlas column
<code>var/gene_name</code>	8,561 unique symbols	Display a readable label
<code>var/gene_id</code>	8,563 Ensembl gene IDs	Reconcile gene features across sources

There are fewer distinct display symbols than columns. The unique index and gene IDs preserve the column distinctions even where the displayed names repeat.

7.7 Construction of the 5,000-gene response axis

The GEARS response surface selects 5,000 positions from the relevant gene measurements. The axis builder reconciles the GEARS gene table with the atlas through Ensembl gene IDs and retains the corresponding atlas column.

For each position, the output table records the raw GEARS symbol and Ensembl ID, the matched atlas symbol and Ensembl ID, the atlas unique column key, and the final resolved_gene_key.

Let $Y = \{0, \dots, 4999\}$ be the response positions and K the set of unique resolved key strings. The mapping $k_Y: Y \rightarrow K$ assigns one key to each position. The metadata mappings $s_Y(y)$ and $e_Y(y)$ give the display symbol and Ensembl ID at position y .

Uniqueness applies to k_Y : different positions have different resolved keys. A display symbol in s_Y may repeat. This lets reporting retain the source labels while model access uses unambiguous keys.

7.8 TBCE: duplicate display symbol, distinct response coordinates

The reported response axis contains two coordinates labeled TBCE [24]:

Axis index	Raw symbol	Ensembl gene ID	Resolved key
535	TBCE	ENSG00000285053	TBCE
536	TBCE	ENSG00000284770	TBCE-1

The Ensembl IDs distinguish the gene records. The local keys distinguish the model positions. Selecting TBCE-1 retrieves the second position while a report may still display its raw symbol as TBCE.

The suffix -1 is a MANTRA uniqueness convention. It is neither a biological isoform designation nor an external database version. Its interpretation comes from the response-axis contract.

7.9 The two axes as distinct types

Use two types of model key:

```
PerturbationRowKey = source experimental symbol in split order
ResponseColumnKey = unique resolved_gene_key in GEARS order
```

A prediction matrix of shape [273, 5000] has 273 hold-perturbation rows and 5,000 response columns. Entry (i, j) predicts response gene j for intervention i. Its row label and column key are supplied by separate records.

The target gene may lie outside the selected response set. A response display symbol may also repeat or use a different naming snapshot. Keeping the two types separate makes those cases explicit.

7.10 Identifier flow

Both branches start from the atlas and construct different ordered outputs.

7.10.1 Perturbation branch

Read cell metadata; retain the exact perturbation symbol; locate it in fit, tune, or hold order; resolve it in the pinned HGNC snapshot; attach Ensembl and NCBI cross-references; select biological inputs under the locus-type policy; align the resulting data to the model's perturbation row.

7.10.2 Response branch

Read the atlas var table and GEARS axis; reconcile Ensembl gene IDs; retain the corresponding atlas column keys; assign 5,000 unique resolved_gene_key values; write the model columns in GEARS order.

A crosswalk may connect a target record with a response record. Keep that mapping and its source separate from either axis's ordering.

7.11 Required resolved identity record

A source-specific biological-input row carries the following fields where they apply. The table continues on the next page.

Field	Meaning
perturbation_identity	Exact source intervention-row label
match_kind	Relation used for resolution: approved, previous, alias, stable-ID, control, or explicit status
hgnc_id	Stable nomenclature-record identifier
approved_symbol	Approved symbol in the pinned HGNC snapshot
ensembl_gene_id	Resolved Ensembl gene accession

Field	Meaning
ncbi_gene_id	NCBI GeneID stored as a string
locus_type	Pinned nomenclature or annotation class
raw_transcript_id	Transcript label carried by the experiment
current_transcript_ids	Transcript records in the selected annotation release
uniprot_accessions	Supported protein entries, including an explicit empty result
gencode_release	Annotation release
ensembl_release	Paired Ensembl release
genome_assembly	Reference coordinate system
source_snapshot_sha256	Hash of the exact resolver catalog or annotation source
resolution_provenance	Resolver, version, parameters, timestamp or run ID

The record may be stored in JSON, Parquet, TSV, or another format that preserves its declared types. Define missing-value meanings. Under this document's rule, an empty protein list means resolution completed and found an empty set of supported accessions. A null field may mean that the field was outside the source's scope; the schema must specify that interpretation.

7.12 Resolution precedence

The stated nomenclature policy evaluates matches in this order:

1. Exact approved HGNC symbol.
2. Unique previous-symbol match.
3. Unique alias-symbol match.
4. Direct stable-ID cross-reference supplied by the source.
5. Explicit ambiguous or unresolved result.

Preserve the input and all candidate evidence. A source Ensembl ID provides additional evidence for checking or disambiguating the symbol match. Transcript membership is checked in the pinned annotation release.

7.13 Using Ensembl IDs for locus and matrix reconciliation

In obs/gene_id, the ID describes the cell's perturbation target. In var/gene_id, it describes the measured gene feature. The axis builder uses the shared namespace to match features across the GEARS and atlas tables.

When both roles appear in one record, names such as target_ensembl_gene_id and response_ensembl_gene_id expose the difference. A shared identifier format permits a lookup; the field role determines what the result means.

7.14 Transcript interpretation

Keep the library's raw transcript string and look up its interpretation in GENCODE release 44. Separate the stable accession stem from its version before comparing the old label with the selected release.

Recorded value	Question answered
Raw transcript string	What did the library record?
Namespace and stable stem	Which identifier system and transcript lineage?
Raw feature version	Which version did the source name?
Release lookup result	What does the selected annotation contain?
Current gene membership	Which gene owns the transcript in that release?
Match relation	Exact accession.version, stem continuation, historical context, or absent in release?

An exact version match and a stem-only continuation provide different evidence. Keeping the relation makes annotation changes visible while preserving the original experiment.

7.15 Protein-derived prior coverage

A **protein-derived prior** is information computed from a selected protein sequence and supplied to the model. Its row requires a supported protein product, a resolved accession, and the exact sequence hash.

The selection policy states whether to use reviewed entries, canonical sequences, or a defined isoform set. For ALG1L1P, the reported current-locus policy yields an empty UniProt accession list. Historical or similarity-based protein references remain separate evidence; using them as a protein product requires evidence for that locus.

7.16 Coordinate-derived prior coverage

A **coordinate-derived prior** summarizes measurements or properties over genomic intervals. Chromatin signals, regulatory measurements, and genomic-neighborhood features require the interval definition as well as the measured source.

Record the assay or file accession, source assembly, annotation release, aggregation window, strand policy, any lift-over operation, exact input hashes, and the meaning of the output feature.

For example, the ENCODE file supplies signal values while the GENCODE GTF supplies the promoter interval over which they are summarized. Their common reference coordinates connect the inputs.

7.17 Project-local identifiers

MANTRA defines local keys for two domains: intervention rows and response coordinates. The table on the next page identifies their separate uses.

Local namespace	Domain	Example
perturbation identity	Ordered experimental intervention rows	ALG1L
resolved gene key	Ordered response-gene coordinates	TBCE-1

The project pins the data and mappings that define these keys. A new resolver snapshot can update external metadata while retaining the established row and column order. Any intentional axis change requires its own versioned contract.

7.18 Source artifact identity

An **artifact** is a stored input or output, such as a matrix, mapping table, or sequence file. To use it correctly, a consumer needs both a description of its bytes and a description of their meaning.

Technical description	What it establishes
Path or content-addressed location	Where to retrieve the artifact
SHA-256 and byte count	Which file was retrieved
Shape and dtype	Array dimensions and stored value type
Required keys and fields	Expected data structure
Producing script and command	How the artifact was generated

Meaning and use	What it establishes
Meaning of X	Counts, log1p CP10K, residuals, or another declared quantity
Default cell mask	Which observations define the standard view
Identifier-field meanings	What each label or accession names
Canonical axis order	Which object each row and column represents
Held-out content	Whether targets or labels restricted by the benchmark are present
Permitted uses	Which downstream operations the contract allows

The **build report** records how the file was produced. The **manifest** states how a consumer must interpret and use it. The technical and interpretive checks work together: a matching hash identifies the expected file, while the manifest identifies the quantity and axes the model is reading.

8 Worked Identity Case Studies

8.1 Case study 1: ALG1L across experimental and current contexts

The reported atlas stores this target metadata [24]:

```
perturbation label: ALG1L
gene ID:           ENSG00000189366
transcript:        ENST00000340333.3
```

The pinned HGNC snapshot resolves ALG1L as a previous symbol for ALG1L1P, record HGNC:33721. That record links to Ensembl gene ENSG00000189366.

GENCODE release 44 represents the gene as ENSG00000189366.13, a transcribed pseudogene named ALG1L1P, and supplies representative transcript ENST00000611639.4. In the same release, transcript stem ENST00000340333 appears at version .7 under ENSG00000291096.1, a neighboring lncRNA record.

The old transcript label and the release-44 transcript therefore describe different annotation contexts. A stem match alone would hide the changed membership. Keep the original gene/transcript pairing as library metadata and record the later lookup separately.

Layer	Identity retained
Benchmark row	ALG1L
Source target gene	ENSG00000189366
Source transcript label	ENST00000340333.3
HGNC record in the snapshot	HGNC:33721
Approved symbol in the snapshot	ALG1L1P
GENCODE 44 gene feature	ENSG00000189366.13
GENCODE 44 representative transcript	ENST00000611639.4
Protein coverage	Empty under the project's current protein-coding criterion

The resulting record preserves the experiment, explains the naming update, exposes the transcript change, and states the protein-coverage result.

8.2 Case study 2: TBCE and response-axis uniqueness

The GEARS response surface contains two coordinates displayed as TBCE. Their Ensembl gene IDs differ. The axis builder keeps both and assigns the local keys TBCE and TBCE-1, as shown in §7.8.

The model predicts each coordinate separately. Array access uses its integer position or unique resolved key. A report displays the raw symbol together with the Ensembl ID so that the reader can distinguish the two records.

Repeated display labels may arise from source construction, naming history, or annotation choices. The remedy is to preserve the source labels and add an explicit unique coordinate key. The source label remains available for interpretation.

8.3 Case study 3: TP53 across gene, transcript, sequence, and protein namespaces

TP53 illustrates how one protein-coding case uses several identifier systems:

Resource	Example object
HGNC	HGNC:11998, approved symbol TP53
Ensembl gene	ENSG00000141510
Ensembl transcript	The selected ENST feature among the gene's transcript records
RefSeq RNA	NM_000546.6
RefSeq protein	NP_000537.3
UniProt	Reviewed entry P04637 and represented isoforms
GENCODE	Transcript structures and tags in a specified GRCh38 annotation release
ENCODE	Experiments measuring RNA, chromatin, or protein binding at the locus

Choose the identifier for the task. Use HGNC for approved naming, Ensembl and GENCODE for genomic structure, RefSeq for the specified reference sequence, and UniProt for protein knowledge. The experiment may cite all of them because each contributes different information.

8.4 Case study 4: a long noncoding RNA locus

Consider a hypothetical lncRNA locus with an HGNC record, an Ensembl gene, several transcript models, and RefSeq noncoding RNA records. Suppose its selected annotation lacks a supported protein product. Its protein mapping is then empty under the project's rule.

The lncRNA still has genomic intervals, promoters, exons, splice junctions, and expression measurements. An experiment can repress its promoter with CRISPRi, measure its RNA, and use chromatin data around the locus. These observations support RNA- and coordinate-based inputs even when the protein-derived input is absent.

The example separates gene identification from protein coverage. Resolve the locus first, then assess each requested source of model information under its own evidence rule.

8.5 Case study 5: a processed pseudogene

Consider a processed pseudogene copied from a mature RNA. Its sequence resembles the parent's joined exons because the RNA intermediate had already undergone splicing. The pseudogene has its own locus, genomic coordinates, Ensembl ID, HGNC symbol, and any evidence of transcription.

A short RNA-derived read matching both parent and pseudogene leaves its source ambiguous. The annotation and counting policy determine how that read is handled. The parent's protein accession belongs to the parent; protein coverage for the pseudogene requires evidence tied to the pseudogene's own record.

Keep parent relationship, locus identity, transcription evidence, and protein coverage in separate fields.

8.6 Case study 6: one dual-guide perturbation record

This illustrative schema shows the parts of a same-gene pair. Ellipses and guide names are illustrative placeholders; measured target coordinates must come from the experiment.

```
perturbation_identity: FNTB
construct_type: same_gene_dual_crispri
guide_A_id: FNTB_guide_A
guide_B_id: FNTB_guide_B
guide_A_target: chr14:...:strand
guide_B_target: chr14:...:strand
raw_transcript_label: ENST...
target_ensembl_gene_id: ENSG...
effector: dCas9-repressor
library_element_id: sgID_AB string
```

Gene-level analysis aggregates valid FNTB cells. Construct-level analysis groups them by exact pair. Guide-level diagnostics inspect A and B individually. The shared source record connects the three analyses.

8.7 Case study 7: one cell from physical experiment to model tensor

Follow a cell in which a delivered dual-guide CRISPRi construct achieves repression. The next page traces that illustrative path from delivery to the model response vector.

1. A lentiviral dual-guide element is delivered to a K562 CRISPRi cell.
2. Its guides direct repression at the intended promoter.
3. The intervention changes the cell's RNA state.
4. Polyadenylated RNA is captured and reverse-transcribed into barcoded cDNA.
5. Sequencing retrieves cell barcodes, UMIs, RNA-derived sequences, and guide-derived sequences from the prepared libraries.
6. Analysis assigns RNA-derived reads to the atlas's 8,563 gene features.
7. Guide analysis calls the paired construct and stores its identity in obs/sgID_AB.
8. The cell receives its gene-level perturbation label in obs/gene.
9. QC and control-matching procedures determine the cell's contribution and target transformation.
10. Aggregation contributes to a response vector on the 5,000-column resolved axis.

The guide identity follows the intervention. The cell identity connects measurements from one partition. Gene identifiers name the measured features. The final model keys preserve the selected order of interventions and responses.

8.8 Case study 8: ENCODE signal summarized to a gene prior

Suppose an ENCODE ChIP-seq track records signal associated with a selected transcription factor on GRCh38. Its ENCFE accession identifies the file; its ENCSR accession identifies the experiment.

To create a gene-level model input, select promoter intervals from GENCODE release 44, summarize the track within the chosen windows, and map the values to the relevant perturbation rows through the gene crosswalk.

Record with the result	Why it is needed
ENCODE experiment and file accessions	Identify the assay and input file
Downloaded file hash and source assembly	Identify exact signal bytes and coordinates
GENCODE release and GTF hash	Identify the annotation used for intervals
Promoter definition and window	Specify which bases were summarized
Aggregation statistic	Specify how signal became one value or feature vector
Gene crosswalk	Connect the annotated locus to the intervention record
Final perturbation row order	Align the result with model inputs

A reader can reproduce the feature only when the measurement, interval definition, summary operation, and output row mapping are all available.

9 Implementation and Governance

9.1 The identity layer as a typed subsystem

The identity subsystem reads source records, parses their identifiers, looks up biological relationships in pinned catalogs, and emits ordered model axes with provenance. The following classes and APIs specify a proposed implementation structure.

Give each identifier an explicit type so that an API requiring a gene record can reject a transcript accession. A possible hierarchy is:

Parent type under Identifier	Specific types
GeneRecordIdentifier	HgncId; EnsemblGeneId; NcbiGeneId
TranscriptIdentifier	EnsemblTranscriptId; RefSeqRnaAccession
ProteinIdentifier	EnsemblProteinId; RefSeqProteinAccession; UniProtAccession
AssayIdentifier	EncodeExperimentAccession; EncodeFileAccession
ExperimentalIdentifier	PerturbationIdentity; GuideIdentity; GuidePairIdentity
ModelCoordinateIdentifier	PerturbationRowKey; ResolvedGeneKey

Dataclasses, Pydantic models, Arrow schemas, or language-specific algebraic types can express these distinctions. The requirement is checked object type at the interface, together with the original identifier value.

9.2 Parsing identifier syntax

Parsing separates the visible parts of an identifier. **Resolution** looks up what the parsed identifier refers to in a catalog. Parsing an ENST prefix establishes a claimed transcript type; resolution establishes whether that transcript exists in the selected release.

```
{
  "namespace": "ensembl",
  "object_type": "transcript",
  "accession": "ENST00000611639",
  "version": 4
}
{
  "namespace": "refseq",
  "object_type": "rna_sequence",
  "accession": "NM_000546",
  "version": 6
}
```

These are separate example objects. Ensembl and RefSeq use the numeric suffix as a version; the UniProt example on the next page uses its suffix as an isoform number.

```
{
  "namespace": "uniprot",
  "object_type": "protein_isoform",
  "accession": "P04150",
  "isoform": 2
}
```

A parser also classifies ENCODE record types from their prefixes. After parsing, resolution consults the pinned catalog for existence, status, relationships, and attributes.

9.3 Resolution result as a first-class object

Return the input, selected record, candidates, supporting evidence, and resolver provenance as one result. This example shows why ALG1L was resolved to ALG1L1P:

```
input_value: ALG1L
input_field: perturbation_identity
input_namespace: experimental_symbol
status: resolved
match_kind: previous_symbol
selected:
  hgnc_id: HGNC:33721
  approved_symbol: ALG1L1P
  ensembl_gene_id: ENSG00000189366
candidates:
  - hgnc_id: HGNC:33721
    evidence:
      - previous_symbol_exact
      - source_ensembl_cross_reference_agreement
resolver:
  catalog_sha256: ...
  code_sha256: ...
  policy_version: hgnc_resolution.v1
```

An ambiguous result retains multiple candidates and an explicit status. The supporting source Ensembl ID is evidence to assess alongside the symbol match.

9.4 Crosswalk tables

A crosswalk row stores a link between two records. Retain source and target namespaces, accessions, and object types; the relation being asserted; both source releases; the evidence source; status or confidence; and the provenance hash.

Relation types include exact match, parent, transcript-of, protein-product-of, previous-symbol-of, alias-of, and MANE match. Status distinguishes exact, unique, multiple, deprecated, and unresolved outcomes. A reader must be able to tell which relationship the row claims and which source supplied it.

A source accession may appear in several crosswalk rows. A transcript analysis selects transcript-of relations in its release; a protein input selects products satisfying its protein policy; a display view selects the approved-symbol relation. Each consumer selects the relationship it needs.

9.5 Ordered axis contracts

An **axis contract** defines which keys belong to an axis and in what order. A proposed perturbation-axis record is:

```
{
  "axis_id": "gears_5k_perturbations.v1",
  "object_type": "experimental_gene_perturbation",
  "key_field": "perturbation_identity",
  "splits": {"fit": 1427, "tune": 357, "hold": 273},
  "source_sha256": "...",
  "ordered_keys_sha256": "..."
}
```

The response-axis contract records 5,000 ordered resolved keys, their Ensembl IDs, source rows, and a hash of the serialized key vector. Matching that hash checks the key values and order under the stated serialization; the axis record supplies their meaning.

9.6 Invariants for the K562 atlas contract

The following checks define the pinned identity surface reported in this document:

1. The raw H5AD hash is 51d6c7ecb349d34ec4coe1bf44a6d832e4113b26f46a319fc7993c028a5f060.
2. Its matrix has 310,385 cell rows and 8,563 expression-gene columns.
3. Cell metadata contain 2,057 gene-target perturbations and one non-targeting control label.
4. The split contains 1,427 fit, 357 tune, and 273 hold rows.
5. Perturbation keys preserve the exact split-contract strings and order.
6. Symbol resolution yields one selected HGNC record or an explicit candidate state.
7. Protein-derived rows retain the selected accession and exact sequence identity.
8. Coordinate-derived rows retain assembly, annotation, and source-file hash.
9. The 5,000 response keys are unique.
10. Ensembl reconciliation retains both TBCE coordinates.

Tests and artifact-registration checks should compare these expectations with the generated records and files. Listing an invariant states the obligation; an executed check supplies evidence that it holds.

9.7 Build reports and manifests

A build report records the producing script, command and parameters, input hashes, output hashes, shapes, dtypes, fields, source-control commit, and resolved configuration. It answers how the artifact was made.

A manifest defines what the artifact represents, its standard fields and masks, the identifier types and ordering of its axes, permitted downstream operations, and benchmark restrictions. It answers how the artifact is to be read and used.

Consumers validate the expected source and structure against the build report, then apply the reading rules in the manifest. These checks connect the actual file with its declared meaning.

9.8 Suggested identity schema

Use one core row per perturbation identity. Store repeated relationships, such as several guides or proteins per perturbation, in linked tables. A **primary key** uniquely identifies a row; a **foreign key** points to a row in another table. These are proposed schemas.

9.8.1 Core perturbation table

Field	Type	Meaning
perturbation_identity	String, primary key	Exact split label
split	Enum	fit, tune, or hold
split_index	Integer	Position within the split
hgnc_id	Nullable string	Selected nomenclature record
approved_symbol	Nullable string	Symbol in the pinned snapshot
match_kind	Enum	Resolution relation
ensembl_gene_id	Nullable string	Selected gene accession stem
ncbi_gene_id	Nullable string	NCBI GeneID
locus_type	Nullable string	Source classification
resolution_status	Enum	resolved, ambiguous, unresolved, control
resolver_provenance_id	Foreign key	Resolver run and snapshot

9.8.2 Raw experimental mapping table

Field	Type	Meaning
cell_or_construct_key	String	Source cell or construct key
perturbation_identity	Foreign key	Benchmark perturbation row
raw_gene_id	String	Source gene ID or control label
raw_transcript	String	Source transcript or library label
guide_A_id	String	Guide in the first position

The remaining fields continue on the next page.

Field	Type	Meaning
guide_B_id	String	Guide in the second position
guide_pair_id	String	Exact sgID_AB
source_file_sha256	String	Raw atlas identity

9.8.3 Protein mapping table

Field	Type	Meaning
perturbation_identity	Foreign key	Project intervention row
uniprot_accession	String	Selected protein entry
isoform	Nullable string	Selected isoform
review_status	Enum	reviewed or unreviewed
sequence_sha256	String	Hash of the declared sequence serialization
mapping_relation	String	Cross-reference path
source_release	String	UniProt release or retrieval snapshot

9.8.4 Response-axis table

Field	Type	Meaning
axis_index	Integer, primary key	Model column position
resolved_gene_key	Unique string	Model-facing response key
raw_gears_symbol	String	GEARS display label
gears_ensembl_gene_id	String	GEARS gene identity
atlas_var_key	String	Exact atlas column key
atlas_gene_symbol	String	Atlas display label
atlas_ensembl_gene_id	String	Reconciled atlas gene identity
mapping_status	Enum	exact, reconciled, or defined source status

9.9 API design

Separate the operations: parse a string, resolve a record, assemble its biological metadata, and construct the response axis. The illustrative calls on the next page retain the proposed API names and inputs. They describe an interface design; execution requires an implementation and the named input objects.

Parsing returns the identifier's parts. Resolution returns candidates and match evidence. Record construction combines the experiment with the pinned catalogs. Axis construction emits the ordered keys, their hash, and diagnostics.

```

parsed = parse_identifier("ENST00000611639.4")
resolution = hgnc_catalog.resolve_symbol(
    symbol="ALG1L",
    supporting_ensembl_gene_id="ENSG00000189366",
)
record = build_perturbation_identity_record(
    perturbation_identity="ALG1L",
    raw_ensembl_gene_id="ENSG00000189366",
    raw_transcript="ENST00000340333.3",
    hgnc_resolution=resolution,
    gencode_catalog=gencode_44,
)
axis = build_response_axis(
    gears_rows=gears_gene_table,
    atlas_var=atlas_var_table,
    join_key="ensembl_gene_id",
)

```

9.10 Diagnostic summaries

Report counts by resolution status and match kind; transcript and protein relationships per record; locus types and protein coverage by type; exact and stem-only transcript matches; duplicated response symbols; guide pairs per perturbation; and cells per construct and perturbation. Also report source releases, assembly coverage, and hashes of every ordered axis.

Emit structured tables alongside the readable report. The tables let a rebuild compare exact rows and counts across runs.

9.11 Versioned policy

A **resolution policy** defines choices such as symbol-match precedence, reviewed-protein preference, handling of changed transcript membership, and local-key construction. Give the policy an identifier, such as `identifier_resolution.v1`.

A policy update produces a new resolver output and comparison report. Keep the source experimental axis fixed unless the project explicitly versions an axis change.

9.12 Comparison between resolver versions

Classify each changed row so that readers can distinguish biological updates from changes to model access:

Change class	Example
Nomenclature	Approved symbol changes; HGNC ID stays the same
Cross-reference	HGNC gains or changes an Ensembl link
Annotation	Transcript set or gene biotype changes
Protein	Reviewed entry or selected sequence changes

Comparison result	Example
Ambiguity change	Alias becomes unique or newly ambiguous
Local-axis check	Perturbation and response keys remain byte-identical

Record the axis check separately from the changed biological fields. This shows whether a metadata refresh also changed the array interface.

9.13 Provenance hashes and content addressing

A cryptographic hash is computed from bytes. Matching hashes provide a practical check that two files have identical bytes under the declared serialization. A **content-addressed** location uses such a hash to identify the stored object.

State what was hashed. Compressed download bytes and decompressed file bytes have different hashes. A sequence cache may hash normalized FASTA, while an axis may hash newline-delimited UTF-8 keys. The normalization and serialization rules must be recorded with the result.

A hash comparison checks content identity. A manifest explains the content's meaning, and validation checks whether the content satisfies that declaration.

9.14 Human-readable documentation and machine-readable contracts

The monograph explains the biological reasoning. An engineering specification defines the schemas, algorithms, and invariants. Artifact-local manifests and build reports record the exact values and procedures for particular files.

Express enforceable requirements in schemas or checks, using the same field names and version identifiers as the prose. JSON Schema, Pydantic, Arrow, YAML contracts, and tests are possible implementations. A written requirement becomes a verified result only after its corresponding check is executed.

9.15 Canonical operational summary

Object	Canonical project identity
Experimental perturbation row	Symbol in split_contract.json, inherited from obs/gene
Raw target locus	obs/gene_id
Raw transcript-design context	obs/transcript
Exact paired intervention	obs/sgID_AB
Human nomenclature	Pinned HGNC record
Genomic annotation	GENCODE 44, Ensembl 110, GRCh38.p14
Protein	Pinned UniProt accession, isoform policy, sequence hash

Object	Canonical project identity
Functional-genomics assay	ENCODE record and exact file accession
Raw expression column	var/_index with var/gene_id metadata
Model response coordinate	Unique resolved_gene_key

To use the identity record, begin with the operation. Reading a benchmark row uses the preserved perturbation key. Selecting a measured column uses its response key or axis position. Looking up biological interpretation follows the relevant cross-reference in a named snapshot.

This division gives each field a defined job. The original experiment remains identifiable, database updates remain traceable, and model arrays retain a verifiable row and column order.

10 Reference Appendices

Appendix A. Mental models

A.1 The biological production chain

A gene's DNA sequence is transcribed into a primary RNA. RNA processing produces a mature transcript structure. A coding transcript supplies the sequence instructions for a protein. The reference assembly and annotation describe where these sequences lie and how their records connect.

A gene may have several transcript models, and protein translation applies to the coding products. Retain the gene, transcript, and protein identifiers at their respective steps.

A.2 The experimental observation chain

A guide construct is delivered to a cell. The experiment captures RNA and guide-derived molecules, prepares barcoded libraries, and sequences them. Analysis assigns the reads to genes and guides, producing a cell-by-gene matrix linked to intervention calls. MANTRA then constructs its intervention and response axes.

Cell barcodes and UMIs identify observations within the assay. Perturbation labels and resolved response keys identify positions in the project data. Source mappings connect the assay records with those project keys.

A.3 The database projection chain

One biological case may link to an HGNC nomenclature record, Ensembl gene and transcript records, GENCODE annotation, an NCBI Gene record, RefSeq sequences, UniProt proteins, and relevant ENCODE experiments. MANTRA's axis keys connect the selected information to model rows and columns.

These are parallel descriptions and linked records. Follow the relationship needed for the analysis and retain the namespace, object type, and release at each endpoint.

Appendix B. Object-level reference table

Object	Identifier	Context retained
Intended intervention row	MANTRA perturbation identity	Split, source atlas, guide construct
sgRNA sequence and vector position	Guide ID	Protospacer, scaffold, assembly, coordinate, A/B position
Dual-guide library element	sgID_AB or construct ID	Both guides, architecture, recombination state
Human nomenclature record	HGNC ID and approved symbol	Snapshot, previous symbols, aliases, locus type

Object	Identifier	Context retained
Annotated gene locus	Ensembl ENSG	Feature version, release, assembly
Annotated RNA structure	Ensembl ENST	Version, release, gene membership, exons
Reference RNA sequence	RefSeq NM_, NR_, XM_, XR_	Sequence version, release
Translated annotation feature	Ensembl ENSP	Version and release
Protein entry or represented isoform	UniProt accession	Isoform, review status, sequence hash, release
Published annotation package	GENCODE release	Ensembl release, assembly, source-file hash
Functional-genomics experiment	ENCODE ENCSR	Sample, target, method, replicates
Downloadable experiment file	ENCODE ENCFF	Format, assembly, pipeline, hash
Raw expression column	var/_index	Symbol and Ensembl gene ID
Model response column	resolved_gene_key	Axis index, symbol, Ensembl gene ID

Appendix C. Prefix and suffix reference

Interpret a suffix using its namespace. A period in an Ensembl or RefSeq accession marks a version; an isoform suffix and a local uniqueness suffix serve different purposes.

Pattern	Meaning
HGNC:<number>	Stable HGNC nomenclature record
ENSG... / ENST...	Ensembl gene / transcript feature
ENSP... / ENSE...	Ensembl protein translation / exon feature
.number after an Ensembl accession	Feature version under Ensembl's rules
NM_... / NR_... / NP_...	RefSeq coding RNA / noncoding RNA / protein
XM_... / XR_... / XP_...	Predicted RefSeq RNA / noncoding RNA / protein
.number after a RefSeq accession	Sequence version
UniProt base accession	Protein knowledgebase entry
-number after a UniProt accession	Represented protein isoform
ENCSR...	ENCODE experiment record

Pattern	Meaning
ENCFF...	ENCODE data file
GRCh38.p14	Human assembly GRCh38, patch release 14
Local suffix in TBCE-1	MANTRA response-coordinate uniqueness key

Appendix D. Guide-pair terminology

Term	Meaning in this document
sgRNA	Single-guide RNA molecule or its sequence design
guide A / guide B	Positions within a paired cassette
single-guide element	One construct encoding one sgRNA
same-gene dual guide	One construct encoding two guides for one gene or promoter context
combinatorial dual guide	One construct encoding guides for two genes or loci
flanking guide pair	Two nuclease guides defining a deletion interval
guide-pair ID	Identifier of the complete A/B construct
direct guide capture	Sequencing guide-derived molecules linked to a cell barcode
guide assignment	An analysis call linking a cell barcode to guide identities
non-targeting control	A control guide designed under the library's targeting criteria
dual index	Two sequencing sample indexes; a separate design choice from guide pairing

Appendix E. Glossary

Accession. A database-assigned identifier for a record. A version suffix identifies a revision where the namespace uses one.

Alias symbol. Another label associated with a gene through literature or database usage, recorded separately from approved and previous symbols.

Annotation. Records identifying features on reference sequences, including genes, transcripts, exons, and coding regions.

Assembly. A reconstructed reference sequence collection with named components and coordinates.

Barcode. A sequence tag linking a molecule to a sample, partition, guide, or other defined experimental unit.

Biotype. A database's classification label for a gene or transcript, such as protein-coding, lncRNA, or processed pseudogene.

Canonical transcript. A representative transcript selected under a particular resource's policy.

cDNA. DNA synthesized from an RNA template by reverse transcriptase. It supplies the DNA intermediate used in the sequencing workflow described here.

Cell barcode. A tag identifying a droplet or cell partition within a library. Library identity distinguishes reused barcodes across datasets.

Chromatin. DNA packaged with histones and other proteins. The packaging influences access to the DNA.

Coding sequence (CDS). The transcript region used to specify the translated amino-acid sequence.

Contig. One continuous assembled sequence, used as a component of a genome assembly.

Cross-reference. A recorded relationship linking one database record to another.

CRISPR activation (CRISPRa). Guide-directed recruitment of transcriptional activators to increase expression.

CRISPR interference (CRISPRi). Guide-directed recruitment of a catalytically inactive Cas effector and repressor to reduce transcription.

Display symbol. A readable gene label used in reports or plots. Repeated labels require separate keys for unambiguous data access.

Dual-guide element. One library construct encoding two sgRNAs in positions A and B.

Enhancer. A regulatory DNA element that increases transcription from a promoter in the relevant cellular context.

Exon. A segment retained in a particular mature transcript. It may include coding sequence, untranslated sequence, or both.

Feature. An annotated object located on a reference, such as a gene, transcript, exon, or coding interval.

Feature version. A revision number attached to an annotation feature under the issuing database's rules.

Gene. In the biological account used here, a DNA region specifying an RNA product. Databases describe genes through locus records and associated transcript models.

Gene model. An annotation record representing a gene locus and its linked transcript structures in a release.

Gene symbol. A short readable label, such as an HGNC-approved symbol or a historical experimental label.

Genome. The complete genetic material of an organism or cell.

GTF. A tab-separated annotation format recording feature coordinates and attributes that connect gene, transcript, and exon records.

Guide RNA. An RNA directing a CRISPR effector to a target. The DNA-targeting systems discussed here use sequence pairing and a compatible PAM.

Guide-pair identity. The exact identifier of a construct containing guides A and B.

Haplotype. A linked set of sequence variants on one chromosome copy.

HGNC ID. Stable identifier of a human gene-nomenclature record.

Intron. A transcribed segment removed during splicing of a particular RNA.

Isoform. A distinct transcript structure or protein sequence associated with a gene; specify which object is meant.

Lift-over. Mapping an interval between reference assemblies using sequence alignments or mapping chains.

lncRNA. Long noncoding RNA: an RNA classified as noncoding and conventionally longer than about 200 nucleotides.

Locus. A location or region in a genome.

MANE. A set of matched RefSeq and Ensembl/GENCODE transcripts with agreeing structure and sequence.

Manifest. A machine-readable description of an artifact's meaning, fields, axes, and permitted uses.

Namespace. The system that issues an identifier and defines how to interpret it.

NCBI GeneID. Stable numeric identifier for an NCBI gene record.

Noncoding RNA. An RNA classified as noncoding because the RNA itself is the annotated product.

Object type. The kind of entity an identifier names: gene record, transcript, protein entry, guide construct, assay file, or model coordinate.

Ortholog. A homologous gene related to another through a speciation event.

Paralog. A homologous gene related to another through a duplication event.

PAM. Protospacer-adjacent motif: DNA sequence recognized by the Cas protein beside the guide target.

Perturbagen. The agent or intervention used to alter the biological system.

Perturbation identity. MANTRA's key for an experimental intervention row: the exact source symbol in split order.

Poly(A) tail. A run of A nucleotides added to an RNA's 3' end and used for oligo-dT capture in the workflow described here.

Primary transcript. The initial RNA product before processing is complete.

Promoter. A DNA region where transcriptional machinery assembles to initiate RNA synthesis.

Protein isoform. A distinct amino-acid sequence associated with the same gene or protein entry.

Pseudobulk. An expression profile obtained by aggregating a specified group of single cells under a stated procedure.

Pseudogene. For protein-derived pseudogenes discussed here, a sequence descended from a protein-coding gene with loss of the ancestral functional protein product.

Release. A published collection of database records and relationships.

Resolved gene key. A unique MANTRA key for one model response coordinate.

Reverse transcription. Synthesis of DNA from an RNA template.

Sequence version. A revision identifier for a sequence, as in RefSeq accession.version.

sgID_AB. The atlas field storing a called paired-guide A/B identity.

Splice junction. The connection between exon sequences in a processed transcript.

Transcript. An RNA produced by transcription; a transcript identifier usually names the represented RNA structure or sequence class.

Transcript isoform. One distinct mature RNA structure associated with a gene.

Transcript model. A database description of an inferred RNA structure and its attributes.

UMI. A molecular tag used to group reads from captured molecules before amplification and estimate molecule counts.

UniProt accession. Identifier of a protein knowledgebase entry; an isoform suffix selects a represented isoform.

Version. A revision identifier interpreted under the issuing namespace's rules.

Appendix F. Evidence inventory for the MANTRA implementation

Artifact	Information it supplies
Raw K562 atlas	obs/gene, obs/gene_id, obs/transcript, obs/gene_transcript, obs/sgID_AB, var/gene_name, var/gene_id
Raw atlas manifest	Shape, X meaning, required metadata, SHA-256
Split contract	Ordered fit, tune, and hold intervention rows
Response gene-axis table	Source symbols, Ensembl IDs, atlas keys, resolved keys
Resolved-key vector	Exact 5,000 response-coordinate keys
HGNC snapshot	Approved, previous, and alias symbols; IDs; cross-references; locus types
GENCODE source manifest	Releases, assembly, source URL, SHA-256
GENCODE GTF	Gene, transcript, exon, CDS, coordinates, biotypes, HGNC attributes

Artifact	Information it supplies
UniProt FASTA cache	Selected accessions and exact sequence bytes
Identifier resolver	Structured result, match kind, candidates, provenance
Sequence coverage report	Gene-to-protein coverage and source identities
Build reports	Scripts, commands, input and output hashes, shapes

Appendix G. Selected references and authoritative resources

1. Bruford EA, Braschi B, Denny P, et al. "Guidelines for human gene nomenclature." *Nature Genetics* 52, 754-758 (2020). DOI: 10.1038/s41588-020-0669-3.
2. HUGO Gene Nomenclature Committee. "Symbol report help" and downloadable field definitions. <https://www.genenames.org/> and <https://hgnc.genenames.org/help/symbol-report/index.html>
3. Ensembl. "Stable IDs: format and versioning." https://mart.ensembl.org/info/genome/stable_ids/index.html
4. Frankish A, Diekhans M, Jungreis I, et al. "GENCODE: reference annotation for the human and mouse genomes in 2023." *Nucleic Acids Research* 51, D942-D949 (2023). DOI: 10.1093/nar/gkac1071.
5. GENCODE. "The Human GENCODE Release History." <https://www.encodegenes.org/human/releases.html>
6. GENCODE. "Gene and transcript biotypes." <https://www.encodegenes.org/pages/biotypes.html>
7. UniProt Consortium. "UniProt: the Universal Protein Knowledgebase in 2025." *Nucleic Acids Research* 53, D609-D617 (2025). DOI: 10.1093/nar/gkae1010.
8. UniProt. "UniProtKB help." <https://www.uniprot.org/help/uniprotkb>
9. NCBI. "NCBI Gene overview and processing." <https://www.ncbi.nlm.nih.gov/datasets/docs/v2/data-processing/gene-processing/gene/>
10. NCBI. "RefSeq Frequently Asked Questions" and "About RefSeq." <https://www.ncbi.nlm.nih.gov/books/NBK50679/> and <https://www.ncbi.nlm.nih.gov/refseq/about/>
11. ENCODE Project. "Data organization" and "Getting started." <https://www.encodeproject.org/help/data-organization/> and <https://www.encodeproject.org/help/getting-started/>
12. scverse. "AnnData" and "On-disk format." <https://anndata.scverse.org/> and <https://anndata.readthedocs.io/en/stable/fileformat-prose.html>
13. Replogle JM, Saunders RA, Pogson AN, et al. "Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq." *Cell* 185, 2559-2575.e28 (2022). DOI: 10.1016/j.cell.2022.05.013.
14. Replogle JM, Bonnar JL, Pogson AN, et al. "Maximizing CRISPRi efficacy and accessibility with dual-sgRNA libraries and optimal effectors." *eLife* 11:e81856 (2023). DOI: 10.7554/eLife.81856.
15. Replogle JM, Norman TM, Xu A, et al. "Combinatorial single-cell CRISPR screens by direct guide RNA capture and targeted sequencing." *Nature Biotechnology* 38, 954-961 (2020). DOI: 10.1038/s41587-020-0470-y.

16. 10x Genomics. "Getting Started: Single Cell 3' Gene Expression." Official workflow documentation. <https://www.10xgenomics.com/support/>
17. 10x Genomics. "Glossary of terms relevant to single-cell products," cell-barcode and UMI entries. <https://www.10xgenomics.com/support/software/cell-ranger/>
18. Mattick JS, Amaral PP, Carninci P, et al. "Long non-coding RNAs: definitions, functions, challenges and recommendations." *Nature Reviews Molecular Cell Biology* 24, 430-447 (2023). DOI: 10.1038/s41580-022-00566-8.
19. Pei B, Sisu C, Frankish A, et al. "The GENCODE pseudogene resource." *Genome Biology* 13:R51 (2012). DOI: 10.1186/gb-2012-13-9-r51.
20. NCBI MeSH. "Reverse Transcription" and "Reverse Transcriptase Polymerase Chain Reaction." <https://www.ncbi.nlm.nih.gov/mesh/68048348> and <https://www.ncbi.nlm.nih.gov/mesh/68020133>
21. NCBI. "What are accession numbers at NCBI?" NLM Customer Support Center. <https://support.nlm.nih.gov/knowledgebase/article/KA-03434/en-us>
22. GENCODE. "The GENCODE Project" and data-access documentation. <https://www.gencodegenes.org/pages/gencode.html> and https://www.gencodegenes.org/pages/data_access.html
23. HUGO Gene Nomenclature Committee. "HGNC Guidelines," including pseudogene and RNA nomenclature. <https://hgnc.genenames.org/about/guidelines/>
24. MANTRA project-local evidence reported by the original monograph: GENE_IDENTIFIER_SYSTEMS_WHITE_PAPER.pdf, raw atlas manifests, split contract, response-gene axis table, GENCODE source manifest, and identifier-resolution reports pinned by the project. Counts and case mappings in this revision retain that source's evidence status.
25. GENCODE. "Format description of GENCODE GTF." https://www.gencodegenes.org/pages/data_format.html. Used to check field descriptions in §3.7.
26. NCBI and EMBL-EBI. "Matched Annotation from NCBI and EMBL-EBI (MANE)." <https://www.ncbi.nlm.nih.gov/refseq/MANE/>. Used to check transcript-matching and representative-selection definitions.
27. Thermo Fisher Scientific. "ViraPower Lentiviral Gateway Expression Kit - FAQs," entry "What is MOI, and how do I know which MOI to use?" <https://www.thermofisher.com/order/catalog/product/K496000/faqs>. MOI is a population-level virus-to-cell ratio; it differs from an observed integration count in a particular cell.
28. Smith LM, Kelleher NL, and the Consortium for Top Down Proteomics. "Proteoform: a single term describing protein complexity." *Nature Methods* 10, 186-187 (2013). DOI: 10.1038/nmeth.2369. Supports the sequence-versus-molecular-form distinction in chapter 2.

Appendix H. Final canonical answer

A consistent identifier system records what each identifier names and keeps the links needed to move between those objects.

For MANTRA, the intervention row uses the exact source symbol in the split contract. The atlas also retains the target gene ID, original transcript label, and called guide-pair identity. These fields preserve the experiment at gene, design, and construct levels.

Pinned biological resources add interpretation. HGNC supplies the human gene's approved naming record. Ensembl and GENCODE identify gene and transcript features in the selected annotation. NCBI Gene and RefSeq provide gene records and reference sequences. UniProt supplies the selected protein record and sequence. ENCODE identifies experimental measurements and files. Coordinates are interpreted on the named assembly, here GRCh38.p14 with GENCODE 44 and Ensembl 110.

The response axis has a separate job. Its 5,000 unique resolved_gene_key values identify model columns in their required order. Ensembl gene IDs connect those positions to the atlas's measured features, while source symbols remain available for display.

Store the source snapshot and mapping procedure with each relationship. Then a reader can follow an intervention to its biological records, inspect the measurements used by the model, and recover the meaning of each row and column.